



ROB 2140 Studio 5000 and FANUC Robotics

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LAB: CREATE NEW ROBOT ETHERNET/IP CONNECTIONS

PROGRAM FILE: FANUC_Robot

OBJECTIVE: Create the Ethernet module and the FANUC communications node.

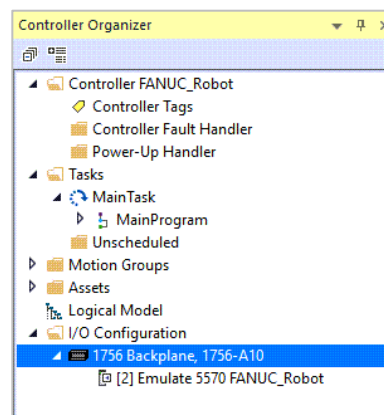
Open the project file.

Filename: FANUC_Robot

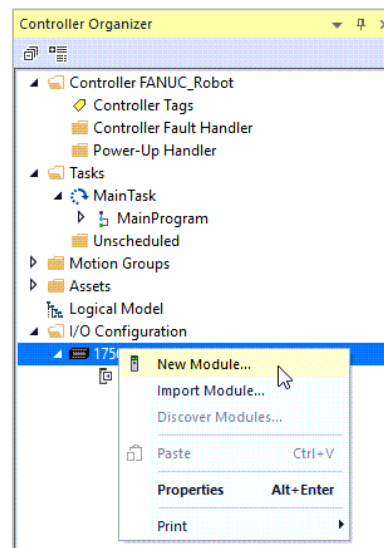
Adding Ethernet/IP Communications Module

Note: different models of processor may have built in Ethernet/IP connections and would not need to be added. The communication properties would still need to be configured.

Right click on **1756 Backplane, 1756-A10**.

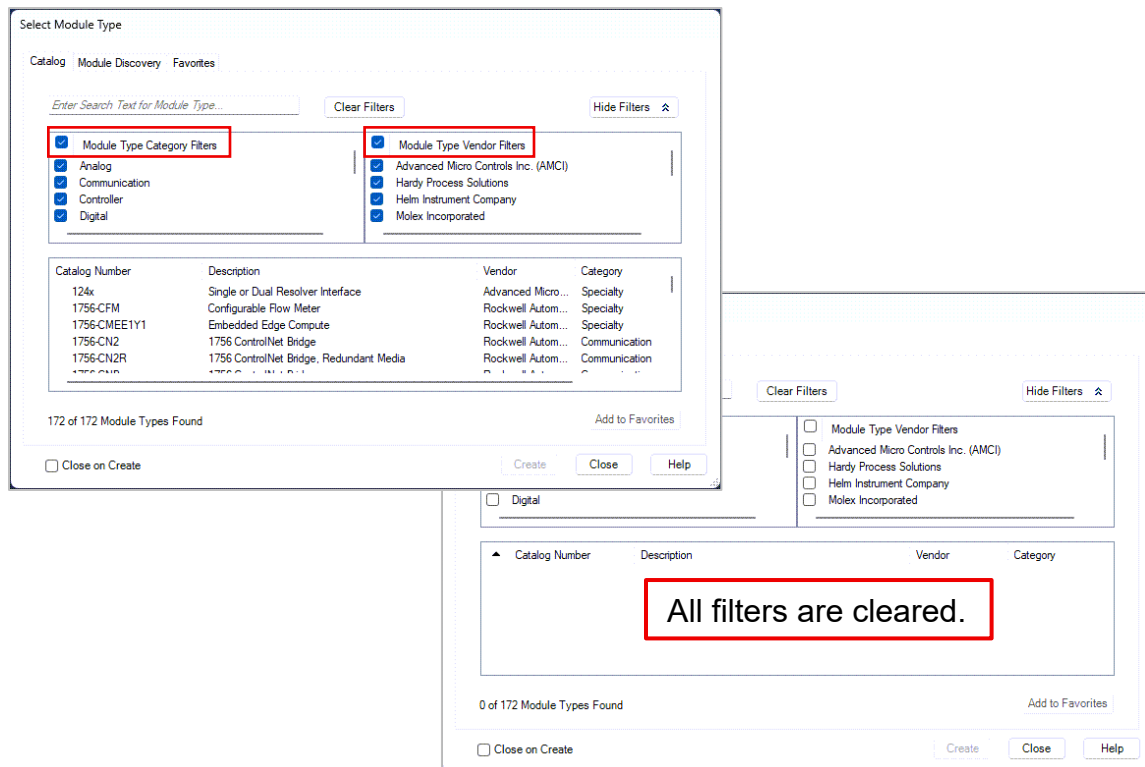


Select **New Module...**



The **Select Module Type** dialog opens.

Uncheck **Module Type Category Filters**.
Uncheck **Module Type Vendor Filters**.



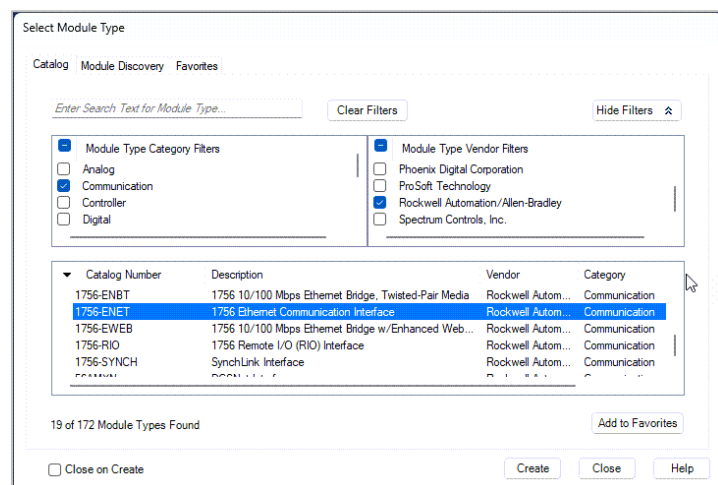
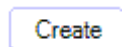
Select the following:

In **Module Type Category Filters** select **Communications**.

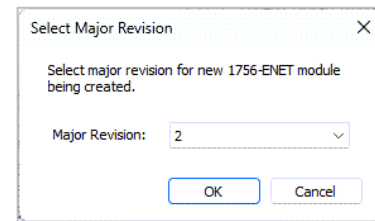
In **Module Type Vendor Filters** scroll down and select **Rockwell Automation/Allen-Bradley**.

In the listing select **1756-ENET** the **1756 Ethernet Communication Interface**.

Select the **Create** button.



Select the **OK** button for the **Select Major Revision**.



The **New Module** dialog opens.

Fill in the following:

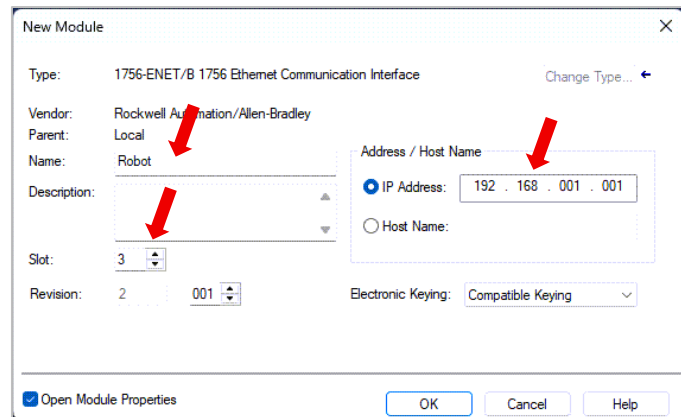
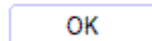
Name: Robot

Slot: 3

IP Address: 192 168 001 001

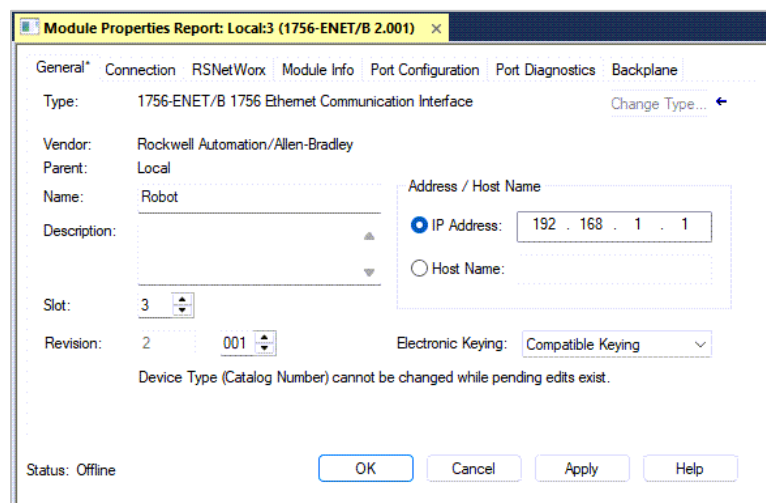
Do not type in spaces

Select the **OK** button.

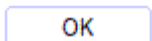


The module's **Properties** dialog opens.

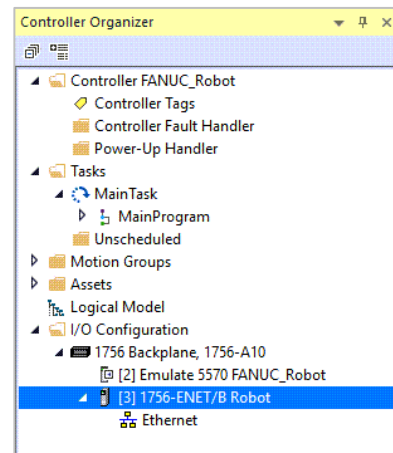
Select the **General** tab and verify that the correct configuration has been entered.



Select the **OK** button.

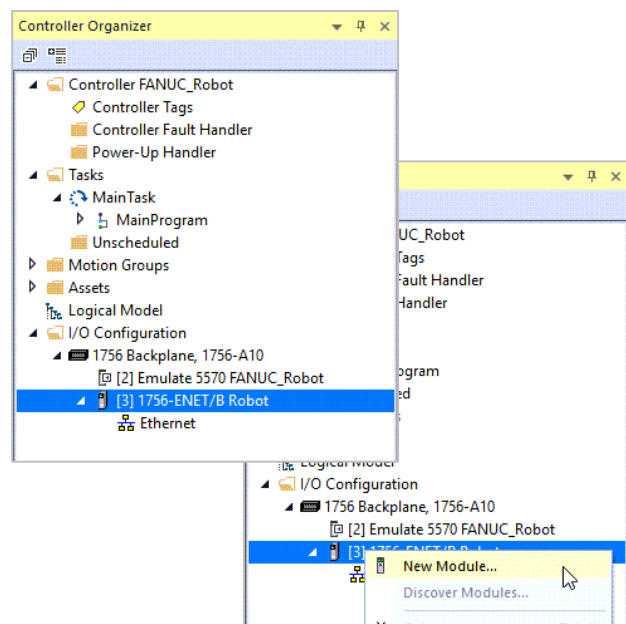


The **1756-ENET** module for Ethernet/IP communications named **Robot** has been added to the **I/O Configuration**.



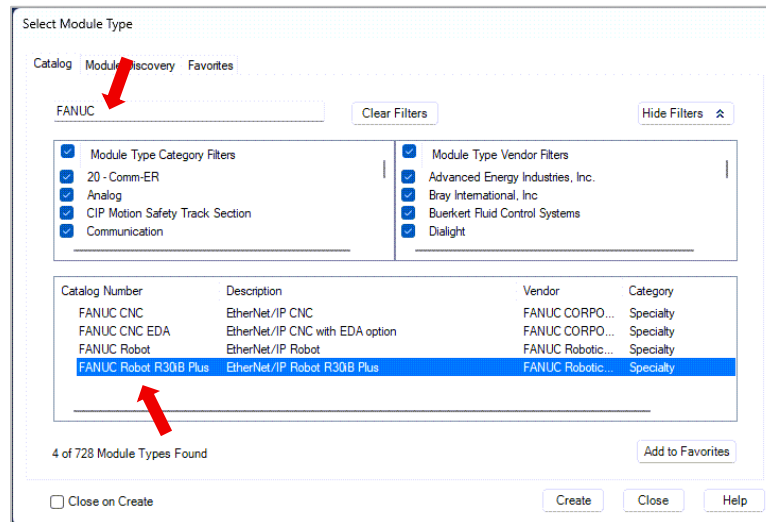
Adding FANUC Communications Node

Right Click on **[3] 1756-ENET/B Robot** and select **New Module**.



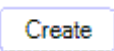
continue

In the Filter field *Enter Search Text for Module Type...* enter **FANUC**.



Select **FANUC Robot R30iB Plus** for Ethernet/IP communications.

Select the **Create** button.



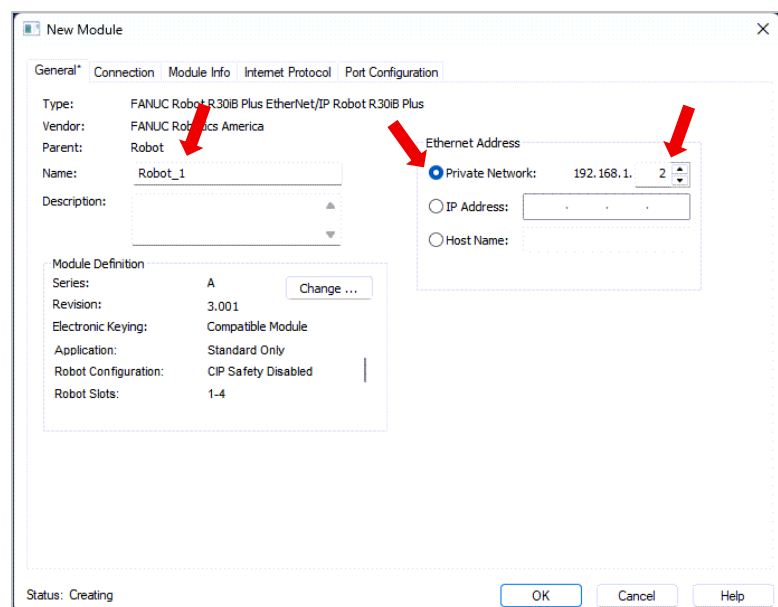
The **New Module** dialog opens.

Change the following:

Name: Robot_1

Select Private Network:

Change IP Address to 2



In the **Module Definition** select the **Change...** button.

Change ...

General Connection Module Info Internet Protocol Port Configur

Type: FANUC Robot R30IB Plus EtherNet/IP Robot R30IB P
Vendor: FANUC Robotics America
Parent: Robot
Name: Robot_1
Description:

Module Definition
Series: A
Revision: 3.001
Electronic Keying: Compatible Module
Application: Standard Only
Robot Configuration: CIP Safety Disabled
Robot Slots: 1-4

Select the **Connection** tab.

Change **Size (8-Bit)** for:
Input to 4
Output to 4

Select the **OK** button.

OK

Module Definition*

Module Connection

EtherNet/IP Standard

Robot Connection(s)	Input	Output
Standard_Slot_01	4	4

Sizes entered here for standard connections are in bytes (8-bits) while sizes in the robot are entered as words (16-bits).

OK Cancel Help

At the prompt for Change Module definition?
select the **Yes** button.

Logix Designer

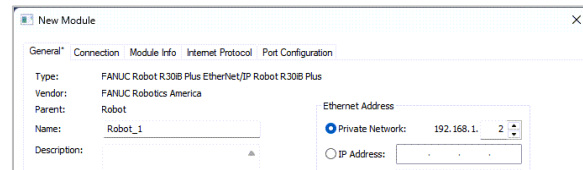
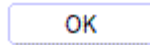
These changes will cause module data types and properties to change. Data will be set to default values unless it can be recovered from the existing module properties. Verify module properties before Applying changes.

Change module definition?

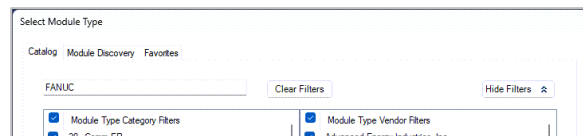
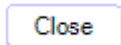
Yes No

Continue

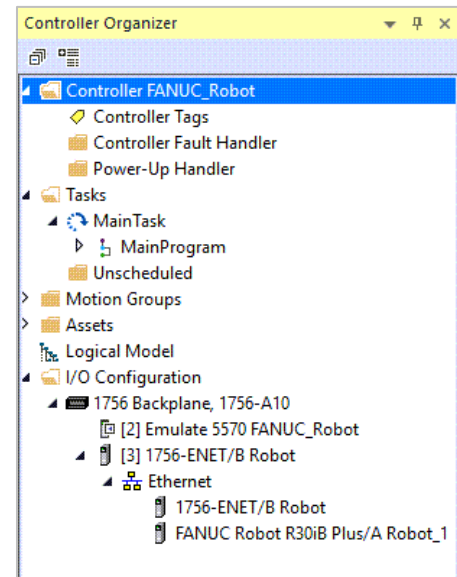
At the New Module select the **OK** button.



At the Select Module Type select the **Close** button.



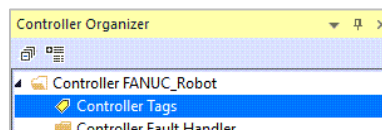
The **FANUC Robot R30iB Plus/A Robot_1** is added to the **Ethernet** connection in the **I/O Configuration**.



Save the PLC program file.

INFORMATION - View the Controller Tag Structure for the FANUC I/O

Open the **Controller Tags**.



The **Robot_1** is listed as **I** (Input) and **O** (Output) as a **Data Type** of **4 Bytes** in an array of **FR:Standard_RobotPlus**.

Name	Alias For	Base Tag	Data Type
▶ Robot_1:I			FR:Standard_RobotPlus_4Bytes:I:0
▶ Robot_1:O			FR:Standard_RobotPlus_4Bytes:O:0

Expand **Robot_1:I** and **Robot_1:O** which are a 4 element array of the **SINT[4]** (Single Integer) data.

Name	Alias For	Base Tag	Data Type
▲ Robot_1:I			FR:Standard_RobotPlus_4Bytes:I:0
▶ Robot_1:I.Input			SINT[4]
▲ Robot_1:O			FR:Standard_RobotPlus_4Bytes:O:0
▶ Robot_1:O.Output			SINT[4]

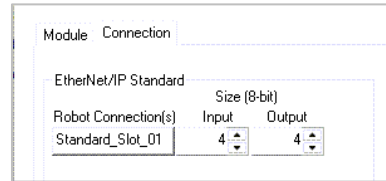
Expand **Robot_1:I.Input** and **Robot_1:O.Output** and the four elements of **SINT** (Single Integer) data words, **[0]** to **[4]**, are shown.

Name	Alias For	Base Tag	Data Type
▲ Robot_1:I			FR:Standard_RobotPlus_4Bytes:I:0
▲ Robot_1:I.Input			SINT[4]
▶ Robot_1:I.Input[0]			SINT
▶ Robot_1:I.Input[1]			SINT
▶ Robot_1:I.Input[2]			SINT
▶ Robot_1:I.Input[3]			SINT
▲ Robot_1:O			FR:Standard_RobotPlus_4Bytes:O:0
▲ Robot_1:O.Output			SINT[4]
▶ Robot_1:O.Output[0]			SINT
▶ Robot_1:O.Output[1]			SINT
▶ Robot_1:O.Output[2]			SINT
▶ Robot_1:O.Output[3]			SINT

continue



The configured **Size (Byte)** of **4** when defining the **Module Definition** configuration.



Expand **Robot_1:I1.Input[0]** and **Robot_1:O1.Output[0]**, the individual **BOOL (Bit)** address **.0** to **.7** are shown.

Examples: **Robot_1:I1.Input[0].2**
Robot_1:O1.Output[0].2

Controller Tags - FANUC_PowerPoint(controller)				
Scope:	FANUC_PowerP	Show:	All Tags	
Name	Alias For	Base Tag	Data Type	
Robot_1:I1			FR:Standard_RobotPlus_4Bytes:I1:0	
Robot_1:I1.Input			SINT[4]	
Robot_1:I1.Input[0]			SINT	
Robot_1:I1.Input[0].0			BOOL	
Robot_1:I1.Input[0].1			BOOL	
Robot_1:I1.Input[0].2			BOOL	
Robot_1:I1.Input[0].3			BOOL	
Robot_1:I1.Input[0].4			BOOL	
Robot_1:I1.Input[0].5			BOOL	
Robot_1:I1.Input[0].6			BOOL	
Robot_1:I1.Input[0].7			BOOL	
Robot_1:I1.Input[1]			SINT	
Robot_1:I1.Input[2]			SINT	
Robot_1:I1.Input[3]			SINT	
Robot_1:O1			FR:Standard_RobotPlus_4Bytes:O1:0	
Robot_1:O1.Output			SINT[4]	
Robot_1:O1.Output[0]			SINT	
Robot_1:O1.Output[0].0			BOOL	
Robot_1:O1.Output[0].1			BOOL	
Robot_1:O1.Output[0].2			BOOL	
Robot_1:O1.Output[0].3			BOOL	
Robot_1:O1.Output[0].4			BOOL	
Robot_1:O1.Output[0].5			BOOL	
Robot_1:O1.Output[0].6			BOOL	
Robot_1:O1.Output[0].7			BOOL	
Robot_1:O1.Output[1]			SINT	
Robot_1:O1.Output[2]			SINT	
Robot_1:O1.Output[3]			SINT	

When expanding **Robot_1:I1.Input[2]** the third word in the array, element **2**, the address for the bit would be **Robot_1:I1.Input[2].3**

Robot_1:I1.Input[2]
Robot_1:I1.Input[2].0
Robot_1:I1.Input[2].1
Robot_1:I1.Input[2].2
Robot_1:I1.Input[2].3



FOR GRADE

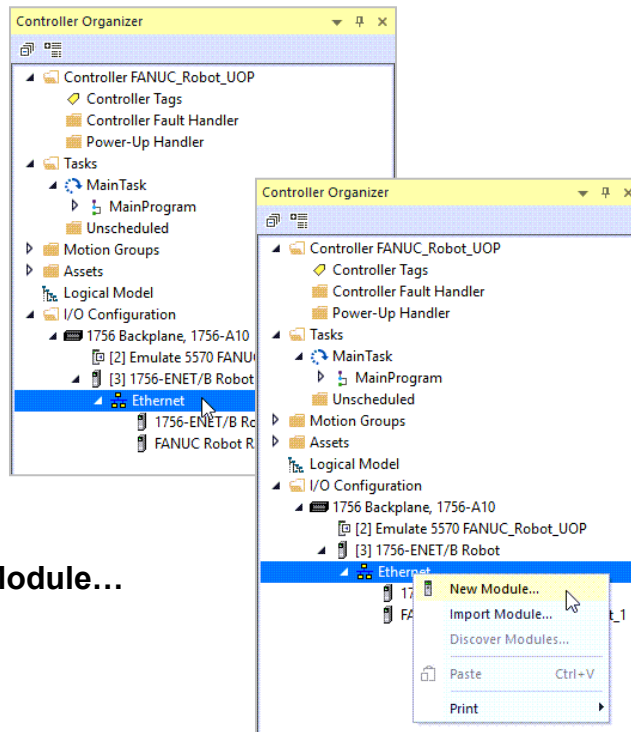
Open Ethernet/IP module **Robot_1 Properties** and show in the General tab the defined settings.

Open the **Module Definition** and show the Input and Output size.

LAB: CREATE NEW MACHINE ETHERNET/IP CONNECTIONS.
PROGRAM FILE: FANUC_Robot_UOP.ACD
OBJECTIVE: Add Ethernet/IP node for Machine for Robot connections.

Right click on **Ethernet**.

Select **New Module...**



Uncheck **Module Type Category Filters** and **Module Type Vendor Filters**.

Select the following:

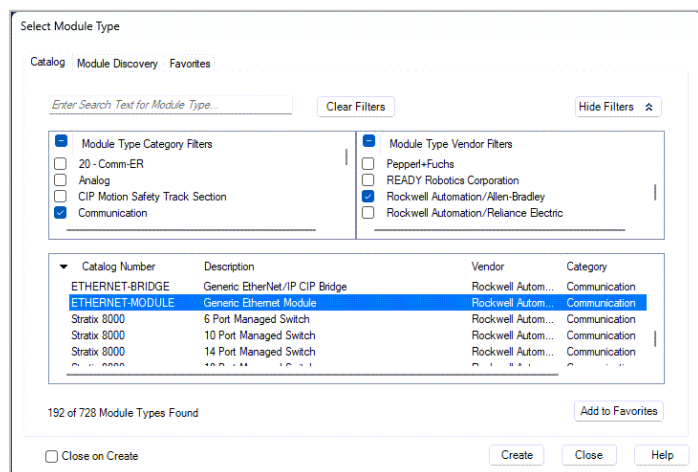
In **Module Type Category Filters** select **Communications**.

In **Module Type Vendor Filters** scroll down and select **Rockwell Automation/Allen-Bradley**.

In the listing select **ETHERNET-MODULE**.

Select the **Create** button.

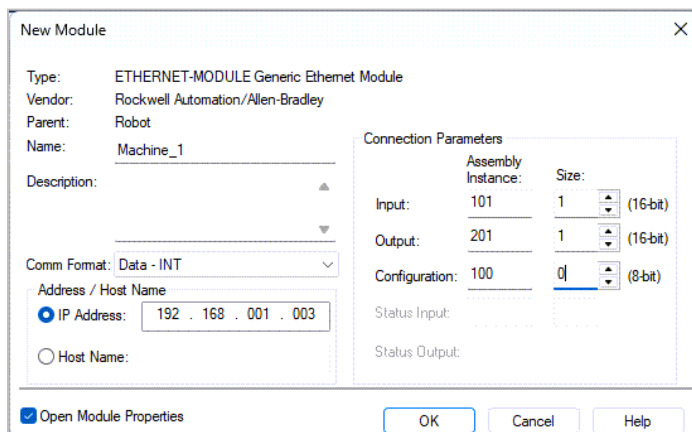
Create



The **New Module** dialog opens.

Change the following:

Name: Machine_1
Comm Format: Data - INT
IP Address: 192 168 001 003
Do not type in spaces
Connection Parameters
Input: 101 **Size:** 1
Output: 201 **Size:** 1
Configuration: 100 **Size:** 0



Select the **OK** button.

OK

The module's **Properties** dialog opens.

Select the **General** tab and verify that the correct configuration has been entered.

Select the **OK** button.

OK

INFORMATION - View the Controller Tag Structure for the Machine

Similar to the Robot, the **Machine_1** is listed as **I** (Input) and **O** (Output) as a **Data Type** of **4 Bytes** in an array of **AB:ETHERNET_MODULE**.

▶ Machine_1:I		AB:ETHERNET_MODULE_INT_2Bytes:I:0
▶ Machine_1:O		AB:ETHERNET_MODULE_INT_2Bytes:O:0
▶ Machine_1:C		AB:ETHERNET_MODULE:C:0

A Configuration array for communications is created.

Expand **Machine_1:I1** and **Machine_1:O1** are a 1 element array of the **INT[1]** (Integer) data.

▲ Machine_1:I		AB:ETHERNET_MODULE_INT_2Bytes:I:0
▶ Machine_1:I.Data		INT[1]
▲ Machine_1:O		AB:ETHERNET_MODULE_INT_2Bytes:O:0
▶ Machine_1:O.Data		INT[1]

continue



Expand **Machine_1:I.Input** and **Machine_1:O1.Output** and one element of INT (Integer) data, **[0]**, are shown.

Machine_1:I		AB:ETHERNET_MODULE_INT_2Bytes:I:0
Machine_1:I.Data		INT[1]
Machine_1:I.Data[0]		INT
Machine_1:O		AB:ETHERNET_MODULE_INT_2Bytes:O:0
Machine_1:O.Data		INT[1]
Machine_1:O.Data[0]		INT

The I/O “On Machine” hardware has 16 Inputs, **.0** to **.15** bit addresses.

*The expanding would show 16 Outputs, **.0** to **.15** bit addresses.*

Machine_1:I		AB:ETHERNET_MODULE_INT_2Bytes:I:0
Machine_1:I.Data		INT[1]
Machine_1:I.Data[0]		INT
Machine_1:I.Data[0].0		BOOL
Machine_1:I.Data[0].1		BOOL
Machine_1:I.Data[0].2		BOOL
Machine_1:I.Data[0].3		BOOL
Machine_1:I.Data[0].4		BOOL
Machine_1:I.Data[0].5		BOOL
Machine_1:I.Data[0].6		BOOL
Machine_1:I.Data[0].7		BOOL
Machine_1:I.Data[0].8		BOOL
Machine_1:I.Data[0].9		BOOL
Machine_1:I.Data[0].10		BOOL
Machine_1:I.Data[0].11		BOOL
Machine_1:I.Data[0].12		BOOL
Machine_1:I.Data[0].13		BOOL
Machine_1:I.Data[0].14		BOOL
Machine_1:I.Data[0].15		BOOL
Machine_1:O		AB:ETHERNET_MODULE_INT_2Bytes:O:0
Machine_1:O.Data		INT[1]
Machine_1:O.Data[0]		INT

FOR GRADE

Show the Tag Editor **Machine_1**, Data Word 2, Bits 0 to 15.

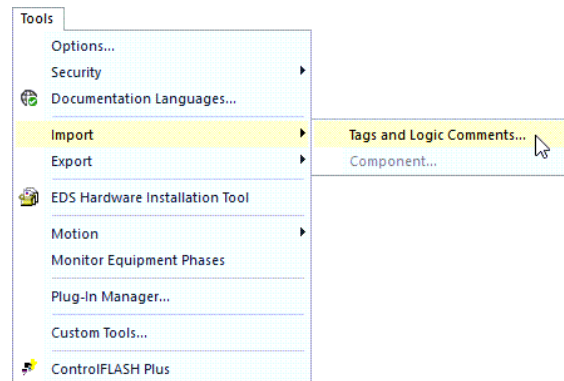


LAB: IMPORT TAGS

PROGRAM FILE: FANUC_Robot

OBJECTIVE: Import the Tag for a .CSV spreadsheet file.

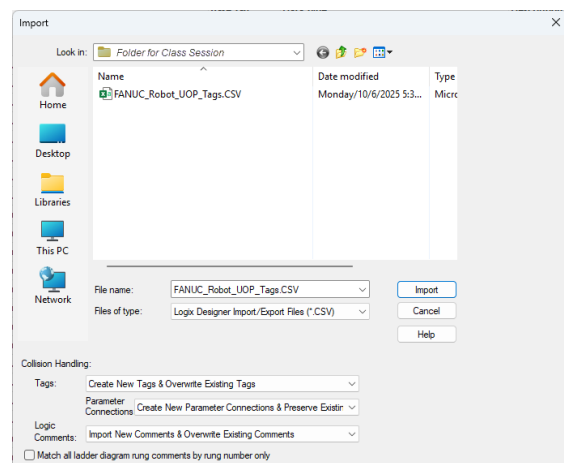
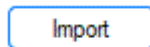
On the Menu Bar select **Tools - Import - Tags and Logic Comments**.



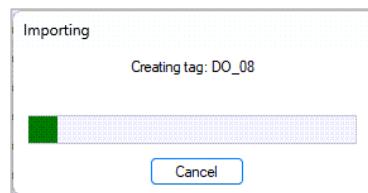
Navigate to the folder for the *Class Session* and **Robot**

Select the file FANUC_Robot_UOP_Tags.CSV

Select the **Import** button.

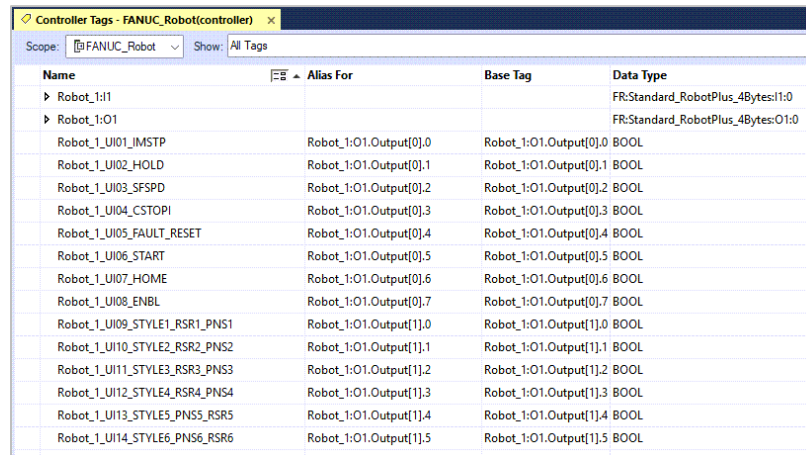


The Import progress is shown.



Continue

The Tags for the **UOP UI** and **UO** (User Operator Panel), **DI** and **DO** (Digital), and **GI** and **GO** (Group) Inputs and Outputs are imported into the application



Name	Alias For	Base Tag	Data Type
Robot_1:1I			FR:Standard_RobotPlus_4Bytes:1:0
Robot_1:O1			FR:Standard_RobotPlus_4Bytes:0:0
Robot_1_UI01_IMSTP	Robot_1:O1.Output[0].0	Robot_1:O1.Output[0].0	BOOL
Robot_1_UI02_HOLD	Robot_1:O1.Output[0].1	Robot_1:O1.Output[0].1	BOOL
Robot_1_UI03_SFSPD	Robot_1:O1.Output[0].2	Robot_1:O1.Output[0].2	BOOL
Robot_1_UI04_CSTOPI	Robot_1:O1.Output[0].3	Robot_1:O1.Output[0].3	BOOL
Robot_1_UI05_FAULT_RESET	Robot_1:O1.Output[0].4	Robot_1:O1.Output[0].4	BOOL
Robot_1_UI06_START	Robot_1:O1.Output[0].5	Robot_1:O1.Output[0].5	BOOL
Robot_1_UI07_HOME	Robot_1:O1.Output[0].6	Robot_1:O1.Output[0].6	BOOL
Robot_1_UI08_ENBL	Robot_1:O1.Output[0].7	Robot_1:O1.Output[0].7	BOOL
Robot_1_UI09_STYLE1_RSR1_PNS1	Robot_1:O1.Output[1].0	Robot_1:O1.Output[1].0	BOOL
Robot_1_UI10_STYLE2_RSR2_PNS2	Robot_1:O1.Output[1].1	Robot_1:O1.Output[1].1	BOOL
Robot_1_UI11_STYLE3_RSR3_PNS3	Robot_1:O1.Output[1].2	Robot_1:O1.Output[1].2	BOOL
Robot_1_UI12_STYLE4_RSR4_PNS4	Robot_1:O1.Output[1].3	Robot_1:O1.Output[1].3	BOOL
Robot_1_UI13_STYLE5_PNS5_RSR5	Robot_1:O1.Output[1].4	Robot_1:O1.Output[1].4	BOOL
Robot_1_UI14_STYLE6_PNS6_RSR6	Robot_1:O1.Output[1].5	Robot_1:O1.Output[1].5	BOOL

Not all Tags will be used in the assignments.



Save the PLC program file.

FOR GRADE

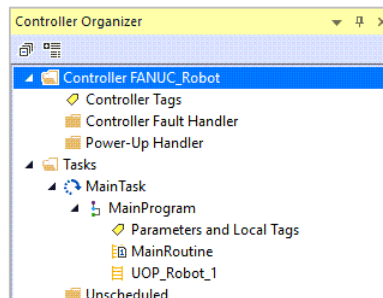
Show the Imported tags.

LAB: ADD REQUIRED USER OPERATOR PANEL SIGNALS

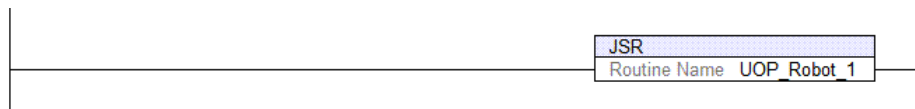
PROGRAM FILE: FANUC_Robot

OBJECTIVE: Add the required UOP outputs for robot operations.

Create a subroutine **UOP_Robot_1**.



In the **Main Routine** call the subroutine **UOP_Robot_1** without any parameters.



In the **UOP_Robot_1** routine add the following Tags to and Output Energize (OTE) instruction.

There are **no input instructions**; the Outputs will be the only instruction on the unconditional rung and use only one output per rung. The Inputs will be added in other lab assignments.

OUTPUT: The Tags were previously imported and **do not need to be created**.

Alias Address	Controller Tag	Description
Robot_1:O1.Output[0].0	Robot_1_UI01_IMSTP	Immediate Stop ⁽¹⁾
Robot_1:O1.Output[0].1	Robot_1_UI02_HOLD	Hold Robot ⁽¹⁾
Robot_1:O1.Output[0].2	Robot_1_UI03_SFSPD	Safety Speed ⁽¹⁾
Robot_1:O1.Output[0].3	Robot_1_UI04_CSTOPI	Controlled Stop ⁽¹⁾
Robot_1:O1.Output[0].4	Robot_1_UI05_FAULT_RESET	Fault Reset ⁽¹⁾
Robot_1:O1.Output[0].7	Robot_1_UI08_ENBL	Enable Motion ⁽¹⁾

All these signals require an ON or Logic 1 for the robot to operate.

⁽¹⁾ See APPENDIX B: FANUC USER OPERATOR PANEL ASSIGNMENTS for complete description.





Save the PLC program file.

FOR GRADE

Show the **UOP_Robot_1** routine.

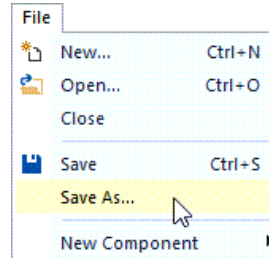


LAB: SAVE AS AN EXPORT FILE

PROGRAM FILE: FANUC_Robot

OBJECTIVE: Export the PLC program as a Library File.

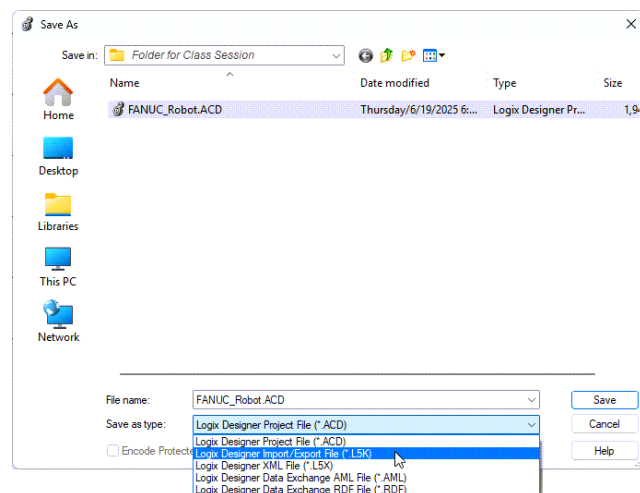
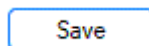
On the Menu Bar select **File - Save As...**



Select the **FANUC_Robot.ACD**

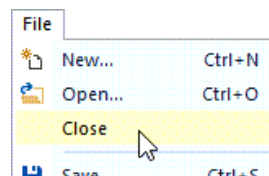
From the **Save as type:** dropdown select **Logix Designer Import/Export File (*.L5K)**.

Select the **Save** button.



Save the PLC program file.

On the Menu Bar select **File - Close Studio**



Use either the **Exit**  icon or Menu bar **File - Exit**.

The **.L5K** file is a text code version of the program's configuration, tags, and logic that may be used to store the project in a **Library** as a **Template** for other PLC program projects.

When a .L5K file is opened the filename must be changed to preserve the original file.

Individual Programs and Subroutine may be exported in the .L5K format. The file may be added to a PLC program by using the Menu bar Tools - Import - Components.



FOR GRADE

Use File Explorer to show the FANUC_Robot.**L5K** file is created.

LAB: IMPORT A PLC FILE

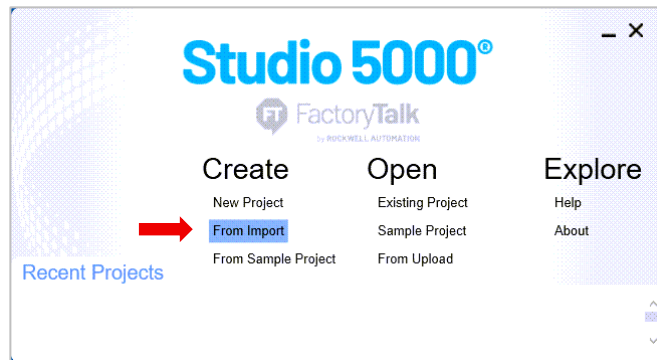
PROGRAM FILE: FANUC_Robot.L5K

OBJECTIVE: Import the PLC program from a Library Template File.

Start **Studio 5000**.

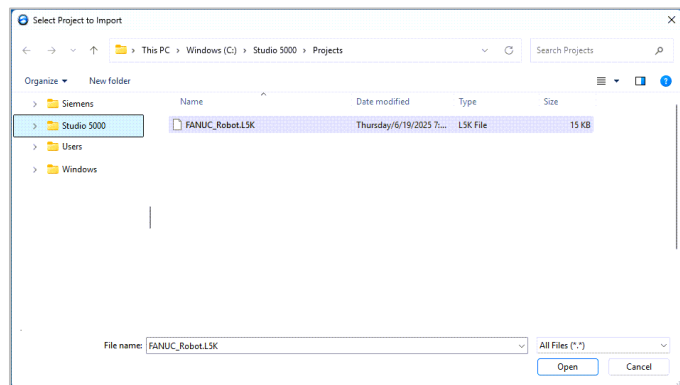
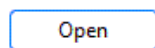


Select **From Import**.



Select the **FANUC_Robot.L5K** file.

Select the **Open** Button.

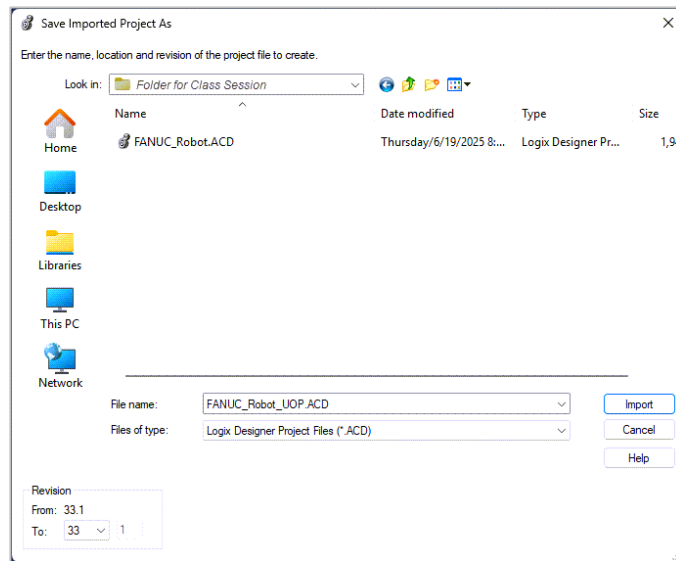
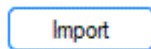


The Studio 5000 software starts.

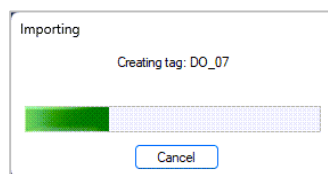


Type the new filename
FANUC_Robot_UOP

Select the **Import** button.

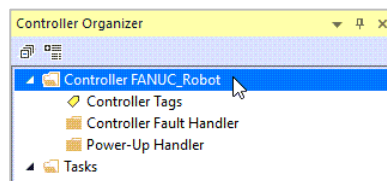


The **Importing** progress bar is shown.

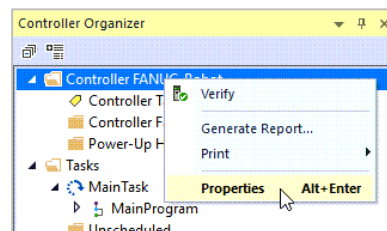


The PLC program opens.

The Controller has the original name and not the UOP project name.



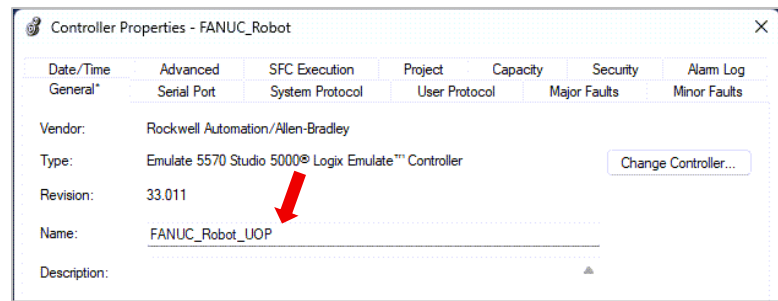
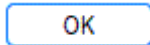
Right click on the Controller and select **Properties**.



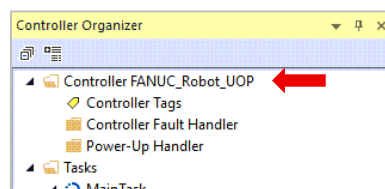
In the Controller Properties change the controller's name.

Name: FANUC_Robot_UOP

Select the **OK** button.



The project is renamed FANUC_Robot_UOP.



FOR GRADE

Show the **FANUC_Robot_UOP** program.

INFORMATION ONLY

The Tag names have "Robot_1" as part of the Tag Names. If the Library .LK5 is used for a different application the Find and Replace function could be used to rename the tags, such as "Robot_2" in the new application.

LAB: PROGRAMMING START AND STOP FOR ROBOT CONTROL.
PROGRAM FILE: FANUC_Robot_UOP.ACD
OBJECTIVE: Programming for the UOP START, IMSTP, and HOLD for a Robot and Machine application..

Create a basic program to control robot operations with User Operator Panel (UOP) signals for START, IMSTP, and HOLD.

UOP Signal	Description
Input: START	When set to ON will cause program execution to begin or resumes a paused program.
Input: IMSTP	The Immediate Stop is a software signal when set to OFF will stop the robot: • Pauses the program execution. • Turns Off power to the servo motors. • Applies the robot's brakes.
Input: HOLD	When set to OFF: • Pauses the program execution. • De-accelerates the robot's motion to a stop.
Output: PAUSED	This output turns ON when a program is paused. If the Teach Pendant or UOP Hold is issued the robot is paused.

The **IMSTP** and **HOLD** are normally a **Logic 1** (ON) signals when the robot is allowed to operate.

A **Logic 0** (OFF) will cause the **IMSTP** to immediately stop the robot and the **HOLD** to pause the robot as described in the table for the signals

The robot is tending a load/unload operation for Machine 1. The PLC based on inputs from machine coordinates the operations with the robot.

The **Machine_1** routine will have the rungs with the **Input Tags** that will be used to control **Output Tags** for the interaction with the UOP signals.

Boolean (Internal) Tags are added as Inputs to the UOP routine rungs for the corresponding control function.

continue



Import the Tag file: FANUC_Robot_UOP_Machine_App_Tags.CSV

Not all Imported Tags will be used in this lab. Unused Tags are for later labs. Only the Tags used are listed.

INPUT:

Alias Address	Controller Tag	Description
Machine_1:I.Data[0].0	Machine_1_MO01_Part_Loading	Loading In Progress
Machine_1:I.Data[0].1	Machine_1_MO02_Part_Unloading	Unloading In Progress
Machine_1:I.Data[0].2	Machine_1_MO03_In_Cycle	In Cycle Production
Machine_1:I.Data[0].3	Machine_1_MO04_Held	Machine Held
Machine_1:I.Data[0].4	Machine_1_MO05_Fault	Machine Faulted

The Loading In Progress or Unloading In Progress or In Cycle Part Production are ON when the machine is actively performing the operation.

The Machine Held and Machine Faulted signal are ON when the machine is Held or Faulted. When either condition is active the corresponding Output turns OFF.

OUTPUT:

Alias Address	Controller Tag	Description
Machine_1:O.Data[0].0	Machine_1_MI01_Part_Load	Load Part Request
Machine_1:O.Data[0].1	Machine_1_MI02_Part_Unload	Unload Part Request
Machine_1:O.Data[0].2	Machine_1_MI03_Part_Prod	Produce Part Request
Machine_1:O.Data[0].3	Machine_1_MI04_Start_Cycle	Start Request
Machine_1:O.Data[0].4	Machine_1_MI05_Hold	Machine 1 Hold Command*

* Logic 1 or ON required for machine operations.

All of the Requests are pulsed ON signals.

The machine will be held when the Hold signal turns OFF.

The Fault is Reset when the signal turns from ON to OFF.

continue



INTERNAL:

Controller Tag	Data Type	Description
Robot_1_Issue_Start_Cycle	Boolean	Start Robot Command
Robot_1_Issue_Hold	Boolean	Hold Robot Command
Machine_1_Issue_Load_Request	Boolean	Load Part Machine Command
Machine_1_Issue_Unload_Request	Boolean	Unload Part Machine Command
Machine_1_Issue_Produce_Part	Boolean	Produce Part Machine Command
Machine_1_Issue_Start_Cycle	Boolean	Start Machine Command
Machine_1_Issue_Hold	Boolean	Hold Machine Command

SUBROUTINES

Create the following subroutine:

Machine_1

MAIN PROGRAM

In the Main Program add an unconditional call for the subroutine **Machine_1** without any parameters.

The Main Program will already have the call to the **UOP_Robot_1** routine.

Machine_1 Routine

Rung 0: Machine_1_MI1_Part_Load

Input Conditions:

not Machine_1_MO02_Part_Unloading **and not** Machine_1_MO03_In_Cycle
and not Machine_1_MO04_Held **and not** Machine_1_MO05_Fault **and**
Machine_1_Issue_Load_Request

Rung 1: Machine_1_MI2_Part_Unload

Input Conditions:

not Machine_1_MO01_Part_Loading **and not** Machine_1_MO03_In_Cycle **and**
not Machine_1_MO04_Held **and not** Machine_1_MO05_Fault **and** Machine_1
_Issue_Unload_Request



Rung 2: **Machine_1_MI3_Part_Prod**

Input Conditions:

not Machine_1_MO01_Part>Loading **and not** Machine_1_MO02_Part
_Unloading **and not** Machine_1_MO03_In_Cycle **and not** Machine_1_MO4
_Held **and not** Machine_1_MO5_Fault **and** Machine_1_Issue_Produce_Part

Rung 3: **Machine_1_MI4_Start_Cycle**

Input Conditions:

not Machine_1_MO03_In_Cycle **and not** Machine_1_MO04_Held **and not**
Machine_1_MO5_Fault **and not** Robot_1_UO04_PAUSED **and** Machine_1
_MI1_Part_Load **or** Machine_1_MI2_Part_Unload **or** Machine_1_MI3
_Part_Prod **and** Machine_1_Issue_Start_Cycle

Rung 4: **Machine_1_MI5_Hold**

Input Conditions:

not Machine_1_Issue_Hold **and not** Robot_1_UO04_PAUSED

INFORMATION ON MACHINE START

The checks for the Machine to Load, Unload, or Produce the Part are in this Machine routine.

Rungs 0 to 2:

The Output signals to Load, Unload, or Produce the Part are dependent on the machine is not In Cycle (this prevents a conflicting command if the machine is already actively loading, unloading, or producing a part). The machine is not Held or Faulted.

The Load and Unload are interlocked to prevent both commands from being issued.

If the above conditions are satisfied, a machine Issue Request is set (ON) for the Load, Unload, or Produce Parts outputs.

Rung 3:

If the machine is commanded to Start if it is not currently In Cycle, and there are no active Machine Hold or Fault, and the Robot is not Paused, and there is a valid Unload, Load or Produce Part (ON), a machine Issue Cycle Start is set (ON).

Rung 4:

If a Machine Hold is issued or Robot is Paused, the rung must be False to cause the machine to halt operations.

continue



UOP_Robot_1 Routine

Drag in a **New Rung 5** for the **UOP START** signal.

Rung 0: Robot_1_UI01_IMSTP

Input Conditions:

not Machine_1_MO05_Fault

Rung 1: Robot_1_UI02_HOLD

Input Conditions:

not Robot_1_Issue_Hold **and not** Machine_1_MI05_Hold

Rung 2: Robot_1_UI03_SFSPD

Input Conditions:

None

Rung 3: Robot_1_UI04_CSTOPI

Input Conditions:

None

Rung 4: Robot_1_UI05_FAULT_RESET

Input Conditions:

None

Rung 5: Robot_1_UI06_START

Input Conditions:

not Machine_1_MO04_Held **and not** Machine_1_MO05_Fault **and not**
Machine_1_MO03_In_Cycle **and** Robot_1_Issue_Start_Cycle

Rung 6: Robot_1_UI08_ENBL

Input Conditions:

None

INFORMATION ON STOP, HOLD, and START

Rung 0: If the machine issues a Fault the robot will Stop and Pause program execution. A fault is a serious problem, and the robot stops immediately and power to the motors is turned Off.

Rung 1: If the machine is Held the robot is Held and Pauses the program. In HOLD the power to the robot's motor will be present, usually for a predetermined time, to allow the robot to continue motion if the hold is for a short time.

Rung 5: A valid robot Start is when there are no Machine Fault or Hold



The checks for a valid Load Part, or Unload Part, or Produce Part are in the Machine routine.

The Machine 1 In Cycle confirms the machine has accepted an operation (Load, Unload, or Produce) command.

A robot Issue Cycle Start is set (ON) to execute the robot operation.

The check for the Machine is then Held or Faulted are included as conditions for the Robot Start because the machine is not operational when those conditions exist. See Rungs 0 and 1 for the signal to cause the robot's motion to be suspended.

FOR GRADE

Show the **UOP_Robot_1** routine and **Machine_1** routine.



LAB: PROGRAMMING FAULT MONITORING AND RESET**PROGRAM FILE:** FANUC_Robot_UOP**OBJECTIVE:** Programming for the UOP FAULT and RESET .

Create a basic program to control robot operations with User Operator Panel (UOP) signals for FAULT and FAULT RESET.

UOP Signal	Description
Input: FAULT_RESET	This input is the external fault reset signal. When this signal transitions from ON to OFF the following will happen: • Error status is cleared • Servo power is turned on • The paused program <i>will not be resumed</i>
Output: FAULT	This output is the error output. This output turns ON when the controller/ program is in an error condition

The **FAULT_RESET** is normally a **Logic 1** (ON) signals when the robot is allowed to operate.

The **FAULT_RESET** must transition from ON to OFF to allow the fault to be cleared.

INPUT:

Alias Address	Controller Tag	Description
Machine_1:I.Data[0].4	Machine_1_MO05_Fault	Machine 1 Faulted

OUTPUT:

Alias Address	Controller Tag	Description
Machine_1:O.Data[0].5	Machine_1_MI06_Fault_Reset	Machine 1 Reset Fault

INTERNALS:

Controller Tag	Data Type	Description
Robot_1_Issue_Fault_Reset	Boolean	Fault Reset Command
Machine_1_Issue_Fault_Reset	Boolean	Fault Reset Command



UOP_Robot_1 Routine

Rung 4: Robot_1_UI05_FAULT_RESET

Input Conditions:

Robot_1_Issue_Fault_Reset

INFORMATION ON FAULT RESET

Rung 4 must go from True (Robot_1_Issue_Fault_Reset ON) to False (Robot_1_Issue_Fault_Reset OFF) for the robot controller to clear the error condition.

Machine_1 Routine

Rung 3: Machine_1_MI04_Start_Cycle

Input Conditions:

Add in series (before parallel branches) Robot_1_UO06_FAULT

The Machine Fault Reset is similar to the Robot; the rung must go from True to False for the error to be cleared.

New Rung 5: Machine_1_MI06_Fault_Reset

Input Conditions:

Machine_1_Issue_Fault_Reset

INFORMATION ON ROBOT FAULT

Rung 3 if the Robot is Faulted a part Load or Unload cannot be performed.

FOR GRADE

Show the changes to the **UOP_Robot_1** routine and **Machine_1** routine.



LAB: PROGRAMMING STATUS FOR READY TO RUN\RUNNING**PROGRAM FILE:** FANUC_Robot_UOP**OBJECTIVE:** Programming for the UOP signals that are needed for a Ready to Run status and Running status .

Create a basic program to control robot operations with User Operator Panel (UOP) signals to determine if the robot is Ready to Run or is Running.

UOP Signal	Description
Output: CMDENBL	This is the command enable output. This output indicates that the robot is in a remote condition and ready to run in AUTO. A Fault condition will turn OFF the output. When SYSRDY is OFF, CMDENBL is OFF. This signal goes ON when the following conditions are all satisfied: • Teach pendant disabled • Mode selection switch is set to AUTO • Robot set to Remote • *IMSTP input is ON • *HOLD input is ON • *SFSPD input is ON • *CSTOPI input is ON • **ENBL input is ON
Output: SYSRDY	This output is the system ready output. This output indicates that the system can provide power for the servo motors.
Output: PROGRUN	This output turns on when a program is running.

INTERNALS:

Controller Tag	Data Type	Description
Robot_1_Status_Ready_Auto	Boolean	Fault Reset Command
Robot_1_Status_System_Ready	Boolean	Fault Reset Requested
Robot_1_Status_Running_Program	Boolean	One Shot Array for Robot



UOP_Robot_1 Routine

New Rung 7: Robot_1_Status_Ready_Auto

Input Conditions:

Robot_1_UO01_CMDENBL

New Rung 8: Robot_1_Status_System_Ready

Input Conditions:

Robot_1_UO02_SYSRDY

New Rung 9: Robot_1_Status_Running_Program

Input Conditions:

Robot_1_UO03_PROGRUN

INFORMATION ON ROBOT STATUS SIGNALS

The **UOP Signal**, [see Table](#), could be used directly for the HMI or other PLC logic. The UOP names are not descriptive for someone who does not know what the signals are indicating. By assigning the UOP signals to PLC Tags a more descriptive name is used.

As an alternative to the added rungs, an **Alias** for UOP Signal could be set up in the Tag Editor.

Name	Alias For	Base Tag
Robot_1_Status_Ready_Auto	Robot_1_UO01_CMDENBL	Robot_1:1.Input[0].0
Robot_1_Status_Running_Program	Robot_1_UO03_PROGRUN	Robot_1:1.Input[0].2
Robot_1_Status_System_Ready	Robot_1_UO02_SYSRDY	Robot_1:1.Input[0].1

FOR GRADE

Show the changes to the **UOP_Robot_1** routine and **Machine_1** routine.



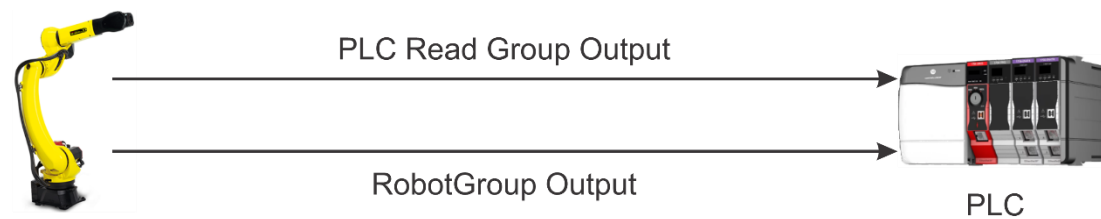
LAB: PROGRAMMING FOR ROBOT GROUP I/O

PROGRAM FILE: FANUC_Robot_Group_IO

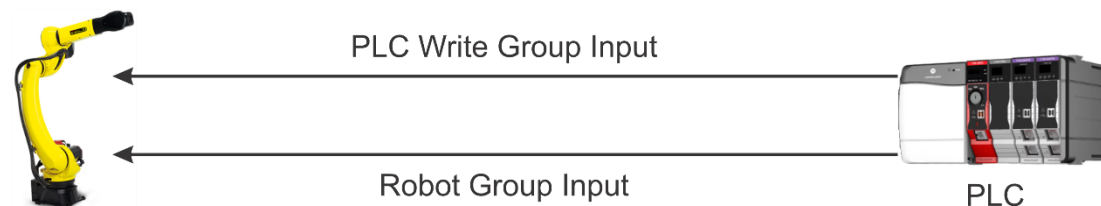
OBJECTIVE: Program PLC transfer robot Group I/O Binary patterns into PLC Words

Create a program for the Robot's GROUP OUTPUT 1 to Read the number value by the PLC and for the PLC to Write a number value to the Robot's GROUP INPUT 1.

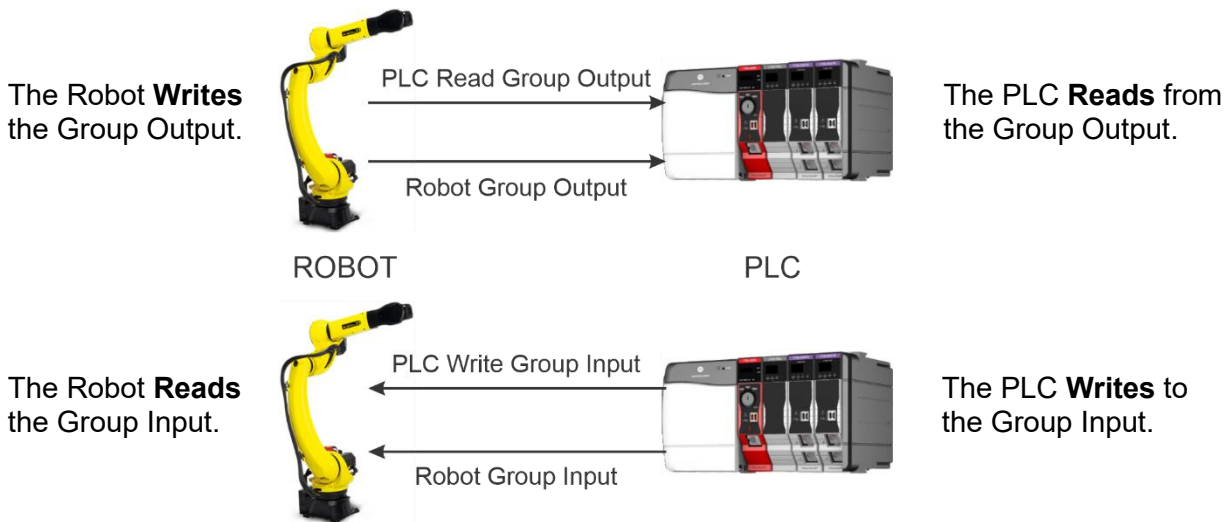
A single discrete (Off-On) signal for the Read and a signal for Write initiate the transactions.



- 1 The robot turns On its Digital Output **DO[1]** to signal for the PLC to Read the Group Output.
 - 2 The robot Output is received by the PLC Input, **Robot_1:I1.Input[2].4**
 - 3 The Robot Writes to the PLC the Binary Bit pattern Group Output 1 **GO[1]**.
 - 4 The PLC Reads the Bit pattern into the Tag Word for **Robot_1:I1.Input[n]** and then places Bit pattern into a Tag Word to store the number value.
- [n] is the Hardware Word Address assigned to the Group Output.*



- 1 The PLC turns On an Output **Robot_1:O1.Output[2].2** to signal the Robot to Read the Group Input.
 - 2 The PLC Output is received by the Robot Input **DI[1]**.
 - 3 The PLC places the Word with the number value into the Tag Word for **Robot_1:O1.Ouput[n]** to Write to the Robot.
 - 4 The Robot Reads from the PLC the Binary Bit pattern as a number value into Group Input 1 **GI[1]**.
- [n] is the Hardware Word Address assigned to the Group Input.*



NOTE: You must enter *all* the **INPUT**, **OUTPUT**, and **INTERNALS** Tags for this lab in the Controller Tag Editor.

Create the Input and Output **Controller Tag** names the assign **Alias For** the Tags.

INPUT:

Controller Tag	Alias For	Description
Robot_1_CMD_PLC_Read_GO1	Robot_1_DO_01	The signal from the Robot to command the PLC to Read the robot's Binary number from Group Output 1

The Base Tag is the Hardware Address Robot_1:I1.Input[2].4

OUTPUT:

Controller Tag	Alias For	Description
PLC_CMD_Robot_1_Read_GI01	Robot_1_DI_01	The signal from the PLC to command the Robot to Read the Binary number to Group Input 1

The Base Tag is the Hardware Address Robot_1:O1.Output[2].2

continue



Assign to the INTERNALS below the **Hardware Word Level Address** in the **Alias For** column for the two Tag names below for Group Input 1 and Group Output 1.

See **Appendix A** for the hardware addresses for the Group Input and Group Output. See the **INFORMATION** below on the Word Addressing.

The **Address Word** for the Tag **Robot_1_Group_Input_1** will be for the **Hardware Word** assigned to robot's **Group Input 1** four (4) **Bit Addresses**.

The **Address Word** for the Tag **Robot_1_Group_Output_1** will be for the **Hardware Word** assigned to robot's **Group Output 1** four (4) **Bit Addresses**.

INTERNALS:

Controller Tag		Description
PLC_Write_Group_Input_1	To the robot	Data Word - Robot Group Input 1
PLC_Read_Robot_1_Group_Output_1		Data Word - Robot Group Output 1

From the robot

INTERNALS:

Controller Tag	Data Type	Description
Robot_1_GI01_Number To the robot	DINT	The PLC Word containing the number value to Write the Group Input to the robot
PLC_Write_GI01_CMD	BOOL	Signal to command the PLC to Write the Group Input number value
Robot_1_GO01_Number From the robot	DINT	Word to store the Group Output number value Read from the robot

INFORMATION – BIT FIELD DISTRIBUTE INSTRUCTION

The **Bit Field Distribute (BTD)** instruction copies the bit number of specified in **Length** from the **Source Word** and **Source Bit** position and distributes to the **Destination Word** and **Dest Bit** position.

BTD	
Source	??
Source Bit	??
Dest	??
Dest Bit	??
Length	??

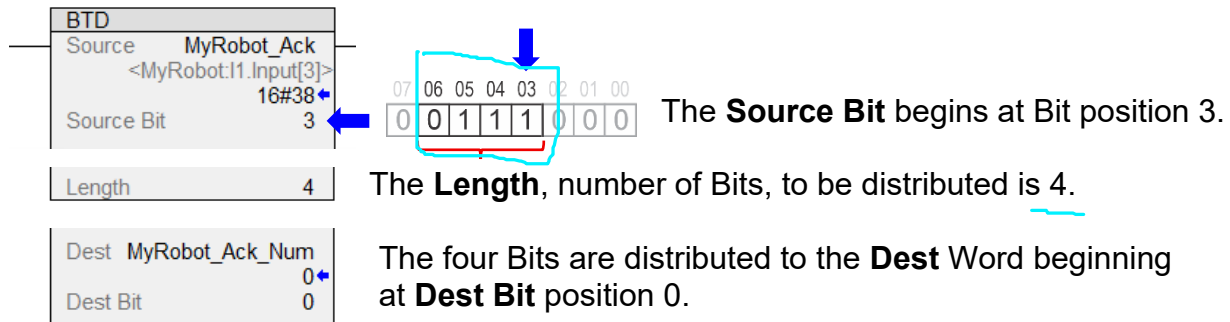
BTD	
Source	MyRobot_Ack
	<MyRobot:I1.Input[3]>
Source Bit	16#38
	3

07	06	05	04	03	02	01	00
0	0	1	1	1	0	0	0
3				8			

In this example the **Source**, a PLC Input, has the Hexadecimal number (16#)38.

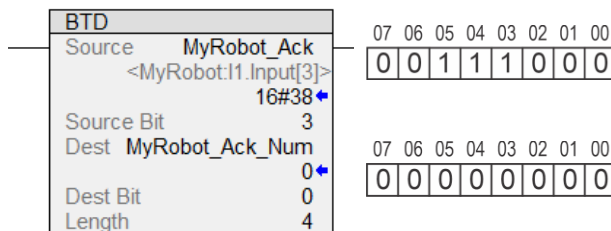
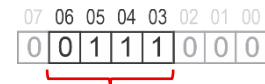
MyRobot:I1.Input[3].3 to MyRobot:I1.Input[3].5 are ON and MyRobot:I1.Input[3].6 is OFF.





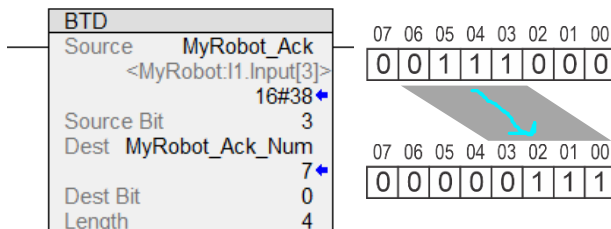
WHEN THE RUNG IS FALSE

In this example, the value of **7**, see the Hex Table on previous page, is moved from the **Source** to the **Destination**.



When the Rung is **False** the shift does not occur.

WHEN THE RUNG IS TRUE



When the Rung is **True** the **Source** Bits **03** to **06** are distributed to the **Destination** Bits **00** to **03**.

Main Routine

Rung 0: (this rung has two Output branches)

When the Internal Tag to **Command the PLC to Write the Group Input is set** (Logic 1) then:

PLC_Write_GI01_CMD

PLC_CMD_Robot_1_Read_GI01

On the first branch the PLC sets the output to **Command the Robot to Read the Binary number to Group Input**.

Robot_1_GI-1_Number

On the second branch the Tag for the **Word Containing the Number Value to Write the Group Input** is placed into the Tag **Data Word - Robot Group Output 1**.

PLC_Read_Robot_1_Group_Output_1

continue



Rung 1:

When the signal to command **Robot to command the PLC to Read the Group Output** is set (Logic 1), the Tag **Data Word - Robot Group Output 1** containing the number value is placed into the Tag **Word to Store the Read of the Group Output** number value from the robot.

PLC_Read_Robot_1_Group_Output_1

Robot1_1_GO1_Number

HMI

Download the PLC program and set the PLC to RUN mode.

Start in **FactoryTalk ME Station** and Load the HMI file **FANUC Robot Group IO**.

PLC DATA WORD GROUP INPUT ACTUAL NUMBER VALUE	DECIMAL	BINARY	DECIMAL	BINARY	DECIMAL	BINARY
1	0001	6	0110	11	1011	
2	0010	7	0111	12	1100	
3	0011	8	1000	13	1101	
4	0100	9	1001	14	1110	
5	0101	10	1010	15	1111	

The screen is divided into sections for the Group Input and Group Output.

Group Input:

The Numeric Input is the **Decimal** number value the **PLC** will **Write** to the Robot's Group Input.

When the PLC Command to Write the number value the Indicator for the Command for the **Robot to Read** the Group Input turn's On.

The PLC Data Word, Output Data Type, the PLC Logic Instruction changes the Bits assigned to the Group Input to the **Binary (Bit) Pattern** for the equivalent number value.

Group Output:

The PLC Data Word, Input Data Type, Bits assigned to the Group Output are changed **Binary (Bit) Pattern** to the number value the **Robot Writes** to the PLC.

When the Robot Commands the PLC to Read the Group Output, the Binary Pattern the PLC Logic Instruction places the equivalent Decimal number value into the Data Word to store the Group Output.



SET GROUP INPUT - PLC OUTPUT DATA WORD

Use the Numeric Input to enter a value between 1 and 15 on the HMI Number Keypad for the ROBOT GROUP INPUT number value.

ROBOT GROUP INPUT
ENTER NUMBER VALUE

9

PLC WRITE TO ROBOT GROUP INPUT

Toggle the switch PLC COMMAND WRITE GROUP INPUT.

PLC COMMAND WRITE
GROUP INPUT



The PLC COMMAND ROBOT READ GROUP INPUT indicator turns On.

PLC COMMAND ROBOT
READ GROUP INPUT



The ROBOT GROUP INPUT Decimal number value is transferred to the PLC DATA WORD GROUP INPUT, Output Data Type **Robot_1:O1.Output[n]**, and changes the Bits assigned to Group Input the Binary (Bit) Pattern

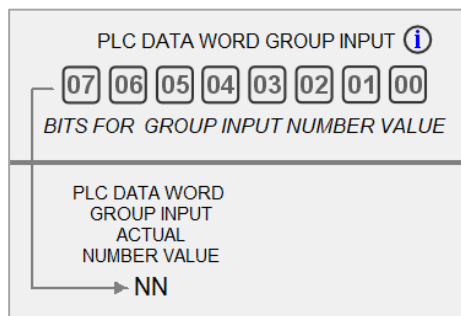
The ACTUAL NUMBER VALUE is the current value at the assigned Group Input Binary (Bit) Pattern.

The graphic does not show the Bit Pattern or Number Value.

This Bit Pattern is the PLC Output Data Word Bits 00 to 07, assigned to the Group Input.

The Bit Pattern of the PLC DATA WORD GROUP INPUT and the ACTUAL NUMBER VALUE should be the same as the ROBOT GROUP INPUT number value entered.

If the values are not the same there is an incorrect parameter(s) in the Logic Instruction.



*This chart is the "weighted value" for the Bit positions, **not** the Bit assignments.*

GROUP INPUT 1 BIT 1 = BINARY 1
GROUP INPUT 1 BIT 2 = BINARY 2
GROUP INPUT 1 BIT 3 = BINARY 4
GROUP INPUT 1 BIT 4 = BINARY 8

See **APPENDIX C: Binary Numbering**

DECIMAL	BINARY
0	0000
1	0001
2	0010
3	0011
4	0100
5	0101
6	0110
7	0111
8	1000
9	1001
10	1010
11	1011
12	1100
13	1101
14	1110
15	1111

continue

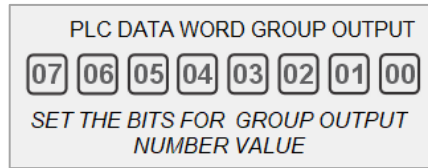


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SET GROUP OUTPUT - PLC INPUT DATA WORD

The graphic shows the Bit Pattern 0.



Set the PLC DATA WORD GROUP OUTPUT Bit Pattern Binary to a number value equivalent to between 1 and 15 Decimal.

This Bit Pattern is the PLC Input Data Word, **Robot_1:I1.Input[n]**, Bits 00 to 07, from the robot's Group Output number value.

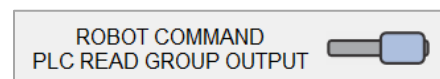
*This chart is the "weighted value" for the Bit positions, **not** the Bit assignments.*

GROUP OUTPUT 1 BIT 1 = BINARY 1
GROUP OUTPUT 1 BIT 2 = BINARY 2
GROUP OUTPUT 1 BIT 3 = BINARY 4
GROUP OUTPUT 1 BIT 4 = BINARY 8

See [APPENDIX C: Binary Numbering](#)

DECIMAL	BINARY
0	0000
1	0001
2	0010
3	0011
4	0100
5	0101
6	0110
7	0111
8	1000
9	1001
10	1010
11	1011
12	1100
13	1101
14	1110
15	1111

PLC READ FROM ROBOT GROUP OUTPUT



Toggle the switch ROBOT COMMAND PLC READ GROUP OUTPUT.

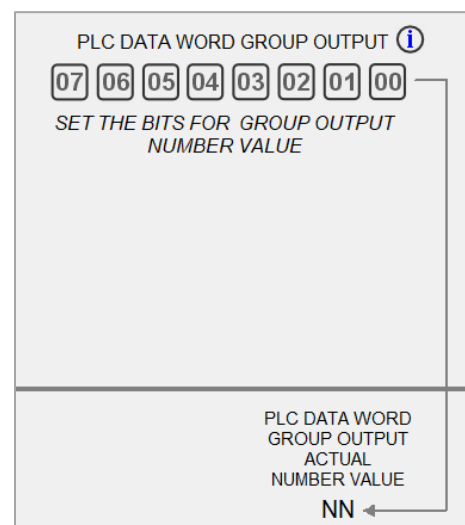
PLC STORED GROUP OUTPUT
NUMBER VALUE

7

The Bittern Pattern of the PLC DATA WORD GROUP OUTPUT should be the same as the PLC STORED GROUP OUTPUT NUMBER VALUE and ACTUAL NUMBER VALUE.

If the values are not the same then:

- *there is an incorrect parameter(s) in the Logic Instruction.*
- *the Bits pressed are not the ones assigned to the Group Output.*



FOR GRADE

Demonstrate the Read and Write of the Group Input and Group Output number values transactions between the PLC and the Robot.



LAB: PROGRAMMING FOR STYLE SELECT

PROGRAM FILE: FANUC_Robot_UOP

OBJECTIVE: Use Style Select signal to designate a program to run and start the program.

Add to the PLC logic the selecting of the program for the robot to execute. The selected program is verified to allow the part load or unload to continue.

UOP Signal	Description
Inputs: STYLE 1-8 RSR 1-8 PNS 1-8	Signals are Binary Numbers read as an Integer value. The inputs select the Style Program Number. The program is selected for execution but does not execute programs . The selected Style Program is executed using the START input or the PROD_START input.
Outputs: STYLEACK 1-8 ACK 1-8 SNO 1-8	Signals are Binary Numbers read as an Integer value. These outputs signal the acknowledgement selected Style Program Number. Normally the acknowledgement UO[11] - UO[18] and the Style number UI[9] - UI[16] are the same number value. A mismatch means an Invalid Program Number Read.
Outputs: SNACK	This output is turned (ON) as a signal to read the Style program number acknowledge outputs.

NOTE: You must enter **all the INTERNALS Tags** for this lab in the Controller Tag Editor.

INTERNALS:

Controller Tag	Data Type	Description
Robot_1_Style_Prog_Number	DINT	Program Style Number
Robot_1_Style_Ack_Number	DINT	Program Style Acknowledge
Robot_1_Style_Prog_Mismatch	BOOL	Style Program Do Not Match
Robot_1_OneShot	DINT[1]	Array for the One Shot Control

continue



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Assign to the **INTERNALs** below the **Hardware Word Level Address** in the **Alias For** column for the two Tag names below for the Style Select Program Number for execution and the Acknowledgement of the Style Select Program Number.

See **Appendix A** for the hardware address. See **INFORMATION** below on the Word Addressing.

The **Address Word** for the Tag **Robot_1_Style_PLC_Output** will be for the Robot UOP Input signals STYLE1, STYLE2, STYLE3, and STYLE4.

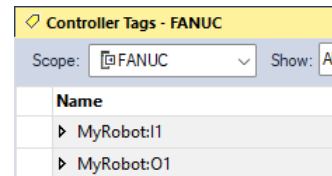
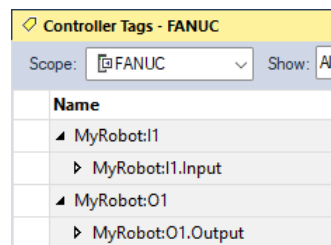
The **Address Word** for the Tag **Robot_1_Style_PLC_Input** will be for the Robot UOP Output signals STYLEACK1, STYLEACK2, STYLEACK3, and STYLEACK4.

INTERNALs:

Controller Tag	Description
Robot_1_Style_PLC_Output	Data Word - Robot Style Program Number
Robot_1_Style_PLC_Input	Data Word - Robot Style Acknowledge

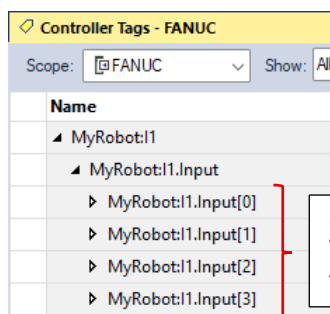
INFORMATION – ADDRESSING AT THE WORD LEVEL

The top level for the Inputs and Outputs defines the robot's **Slot Number**. In this example Slot 1.

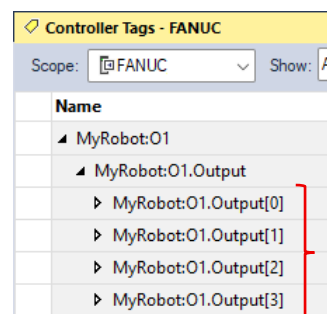


The next level designates the I/O type **Connection**.

Each I/O Type then lists the individual **Words** in the Connection.



Select the Input Word required for the instruction.



Select the Output Word required for the instruction.

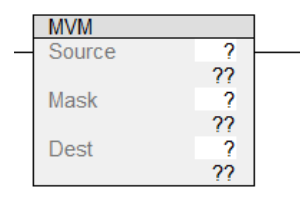
continue



The Tag Name and Values for the Move with Mask, Bit Field Distribute, and Clear instructions below are examples and not what is required for the lab. Except for the MASK.

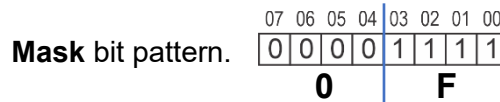
INFORMATION – MOVE WITH MASK INSTRUCTION

The **Move with Mask (MVM)** instruction moves the **Source** word to the **Destination** word bit-by-bit through the **Mask**. word is 8 bits



Mask 1 - Allow change in Destination bit position.
Mask 0 - Leave Destination bit position unchanged.

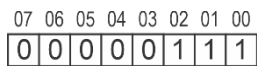
The **Mask** is entered as a Hexadecimal number. Four bits from the Source needs to be moved into the Destination. The Destination is SINT (8 Bit) Data Type, and the **Mask is entered as 16#0F**.



HEX	BINARY
0	0000
1	0001
2	0010
3	0011
4	0100
5	0101
6	0110
7	0111
8	1000
9	1001
A	1010
B	1011
C	1100
D	1101
E	1110
F	1111

WHEN THE RUNG IS FALSE

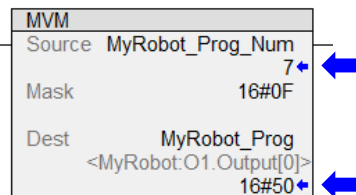
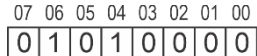
In this example, the value of **7** is moved from the **Source** to the **Destination**. See Hex Table.



The number value at the **Source** is **7**.

Mask 0 0 0 0 1 1 1 1

The number value at the **Destination** is Hex (16#) **50**.



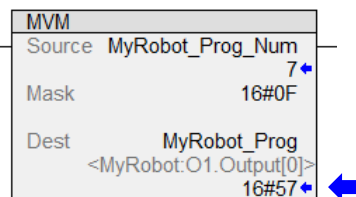
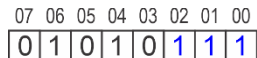
Note:
The Hardware Address by default is Displayed as a Hexadecimal.

WHEN THE RUNG IS TRUE



Only the lower four Bits 00 to 03 were moved.

The number value at the **Destination** is Hex (16#) **57**.



Notice the upper four Bits 04 to 07 are unchanged.

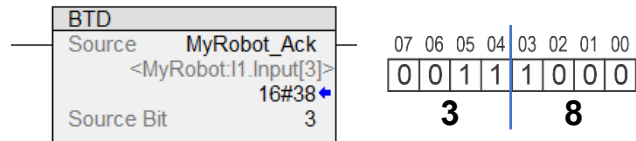
continue



INFORMATION – BIT FIELD DISTRIBUTE INSTRUCTION

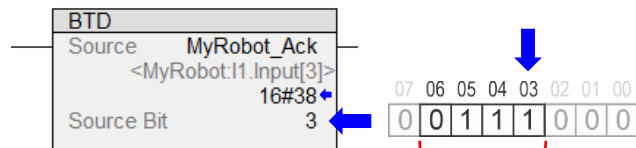
The **Bit Field Distribute (BTD)** instruction copies the bit number of specified in **Length** from the **Source** Word and **Source Bit** position and distributes to the **Destination** Word and **Dest Bit** position.

BTD	
Source	?
Source Bit	??
Dest	?
Dest Bit	??
Length	?



In this example the **Source**, a PLC Input, has the Hexadecimal number (16#)38.

MyRobot:I1.Input[3].3 to MyRobot:I1.Input[3].5 are ON and MyRobot:I1.Input[3].6 is OFF.



The **Source Bit** begins at Bit position 3.

Length	4
--------	---

The **Length**, number of Bits, to be distributed is 4.

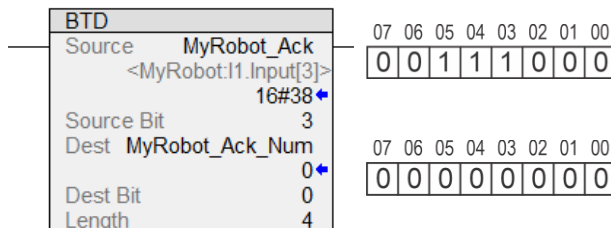
Dest	MyRobot_Ack_Num
Dest Bit	0

The four Bits are distributed to the **Dest** Word beginning at **Dest Bit** position 0.

WHEN THE RUNG IS FALSE

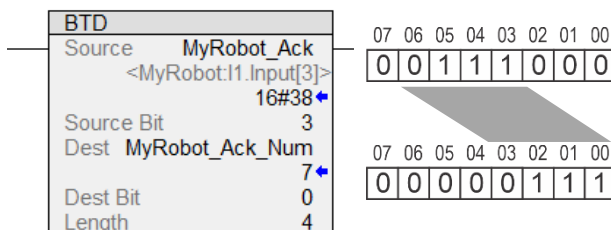
In this example, the value of 7, see the Hex Table on previous page, is moved from the **Source** to the **Destination**.

07	06	05	04	03	02	01	00
0	0	1	1	1	0	0	0



When the Rung is **False** the shift does not occur.

WHEN THE RUNG IS TRUE



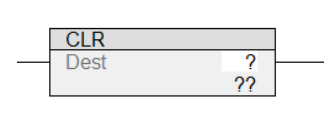
When the Rung is **True** the **Source** Bits 03 to 06 are distributed to the **Destination** Bits 00 to 03.

continue



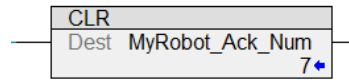
INFORMATION – CLEAR INSTRUCTION

The **Clear (CLR)** instruction zeros out the **Destination word**.



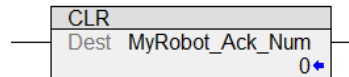
WHEN THE RUNG IS FALSE

When the Rung is False the **Dest** remains unchanged.



WHEN THE RUNG IS TRUE

When the Rung is **True** the **Dest** is zeroed.



SUBROUTINES

Create the following subroutine:

Program_Style_Select_Robot_1

Main Routine

In the Main Program add an unconditional call for the subroutine **Program_Style_Select_Robot_1** without any parameters.

Program_Style_Select_Robot_1 ROUTINE

Rung 0:

When the **UOP Tag to START** the robot is set (Logic 1) the Tag for the **Program Style Number** will be moved into the Tag **Data Word - Robot Style Program Number** for the robot's UOP Style 1 thru Style 4 signals.

Note: only the Bits assigned to the Style 1 thru Style 4 are to change. All other Bits in the Word remain unchanged.

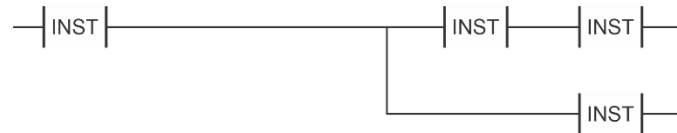
Rung 1: (this rung has two Output branches)

When the robot UOP signal to **Read** the **Style Program Number Acknowledge** is set (Logic 1) then:

The first branch has a One Shot instruction and the instruction to clear the Tag for the **Program Style Acknowledge**.

The second branch shifts the Tag [Data Word - Robot Style Acknowledge](#) for the robot's UOP StyleAck 1 thru StyleAck 4 signals to the Tag for the [Program Style Acknowledge](#).

Note: *the second branch is around the two instructions on the first branch.*



Note: *only the Bits assigned to the Style Ack 1 thru Style Ack 4 are shifted.*

Rung 2:

When the robot UOP signal to [Read the Style Program Number Acknowledge](#) is set (Logic 1) and the [Program Style Number](#) is not equal to the [Program Style Acknowledge](#) the Tag for [Style Program Do Not Match](#) is Latched ON.

Rung : 3

When the robot [Fault Reset Command](#) is issued (Logic 1) the Tag for [Style Program Do Not Match](#) is Unlatched OFF.

UOP_Robot_1 ROUTINE

If the Tag for [Style Program Do Not Match](#) is set (Logic 1) then make the following UOP Output signals go False:

Immediate Stop
Start

Machine_1 ROUTINE

If the Tag for [Style Program Do Not Match](#) is set (Logic 1) then make the following machine Output signals go False:

Start Cycle
Hold

HMI

Download the PLC program and set the PLC to RUN mode.

Start in **FactoryTalk ME Station** and Load the HMI file **FANUC Robot UOP**.

continue



The initial screen has the Robot statues of READY TO RUN IN AUTO and SYSTEM READY.

FANUC Robot User Operators Panel

MACHINE		ROBOT	
START	RESET	START	RESET
HOLD		HOLD	
LOADING PART	OFF	READY TO RUN IN AUTO	ON
UNLOADING PART	OFF	PAUSED	OFF
PRODUCING PART	OFF	SYSTEM READY	ON
		FAULT	OFF
		RUNNING PROGRAM	OFF
		PROGRAM MISMATCH	OFF
MACHINE SYSTEM SIGNAL		PROGRAM STYLE SELECT ENTER PROGRAM NUMBER	
MACHINE START PUSHBUTTON ACKNOWLEDGED		0	
		PROGRAM ACKNOWLEDGED	
		0	

MACHINE	SIMULATED SIGNALS	ROBOT
FAULT	RESET PLC and HMI DATA	FAULT
ISSUE LOAD		
ISSUE UNLOAD		
ISSUE PRODUCE		

PRESET THE BITS FOR STYLE ACKNOWLEDGED NUMBER VALUE

07 06 05 04 03 02 01 00

RESET

SIMULATED SIGNALS

RESET PLC and HMI DATA

The button will reset the **PLC** and **HMI Data**. In most cases this will then allow the PLC program to be tested again. If the problem reoccurs there may be incorrect logic in the PLC program.

MACHINE: ISSUE LOAD PART

In the HMI section **SIMULATED SIGNALS** select the slider for **ISSUE LOAD**.

MACHINE

- FAULT
- ISSUE LOAD
- ISSUE UNLOAD
- ISSUE PRODUCE

ROBOT: SET PROGRAM STYLE SELECT NUMBER

Use the Numeric Input to enter a value between 1 and 15 on the HMI Number Keypad for the **PROGRAM STYLE SELECT** the robot is to **RUN**.

PROGRAM STYLE SELECT
ENTER PROGRAM NUMBER

0

continue



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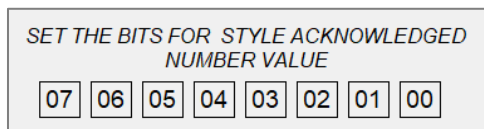
ROBOT: SET PROGRAM STYLE ACKNOWLEDGED NUMBER VALUE

Set the Bits (Blue background is Logic 1) for the Binary Number pattern for the **same number value** as the STYLE SELECT NUMBER used above.

See [Appendix A](#) for the hardware address for Bit positions.

Use the bits assigned to STYLEACK1, STYLEACK2, STYLEACK3, and STYLEACK4.

The graphic shows
the Bit Pattern 0.



DECIMAL	BINARY	DECIMAL	BINARY
0	0000	9	1001
1	0001	10	1010
2	0010	11	1011
3	0011	12	1100
4	0100	13	1101
5	0101	14	1110
6	0110	15	1111
7	0111		
8	1000		

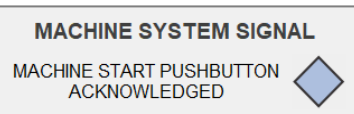
The Machine START must be issued first, followed by the Robot's START.

MACHINE: ISSUE START

Press the Machine START button.

START

The MACHINE START PUSHBUTTON ACKNOWLEDGED
turns On.



INFORMATION - This is a system signal that is set by the machines operating system when a Start has been issued to the machine.

In this application, if there is an active Load or Unload Request the machine is waiting for the coordinating robot Start to begin the selected cycle.

If there is an active Produce Part Request the machine begins the part production cycle.

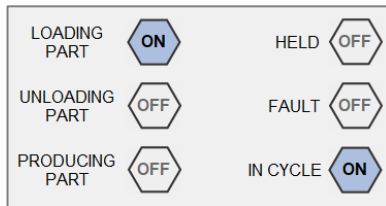


ROBOT: ISSUE START

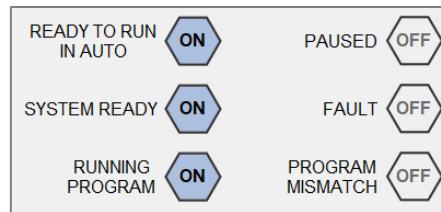
Press the Robot START button.

START

The Machine's LOADING PART and IN CYCLE indicators turn On.



The Robot's RUNNING PROGRAM indicators turn On.



When the Load Part cycle is finished the indicators turn Off.
Note: robot returns to a Paused status.

PROGRAM MISMATCH ON

If there is a robot PROGRAM MISMATCH and FAULT the indicators turn On. The PROGRAM NUMBER and PROGRAM ACKNOWLEDGE NUMBER values are not the same

Press the Robot's RESET pushbutton.

RESET

Set the STYLE ACKNOWLEDGE NUMBER value to the correct Bit pattern for the Program Number.

SET THE BITS FOR STYLE ACKNOWLEDGED
NUMBER VALUE

07 06 05 04 03 02 01 00

Press the Machine START and the Robot START pushbuttons.

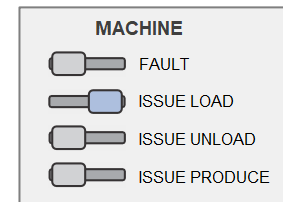
If the problem reoccurs check the PLC logic instruction.

continue



MACHINE: ISSUE PRODUCE PART

In the HMI section SIMULATED SIGNALS select the slider for ISSUE PRODUCE.

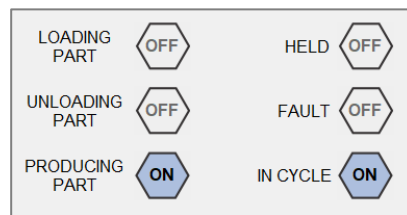


MACHINE: ISSUE START

Press the Machine START button.



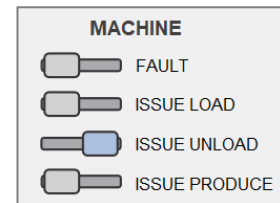
The Machine's LOADING PART and IN CYCLE indicators turn On.



When the Produce Part cycle is finished the indicators turn Off.

MACHINE: ISSUE UNLOAD PART

In the HMI section SIMULATED SIGNALS select the slider for ISSUE UNLOAD.



ROBOT: SET PROGRAM STYLE SELECT NUMBER

Use the Numeric Input to enter (*use a different Program Number*) a value between 1 and 15 on the HMI Number Keypad for the PROGRAM STYLE SELECT the robot is to RUN.



continue

ROBOT: SET PROGRAM STYLE ACKNOWLEDGED NUMBER VALUE

Set the Bits (Blue background is Logic 1) for the Binary Number pattern for the **same number value** as the STYLE SELECT NUMBER used above.

See [Appendix A](#) for the hardware address for Bit positions.

Use the bits assigned to STYLEACK1, STYLEACK2, STYLEACK3, and STYLEACK4.

The graphic shows
the Bit Pattern 0.

SET THE BITS FOR STYLE ACKNOWLEDGED
NUMBER VALUE

07 06 05 04 03 02 01 00

DECIMAL	BINARY	DECIMAL	BINARY
0	0000	9	1001
1	0001	10	1010
2	0010	11	1011
3	0011	12	1100
4	0100	13	1101
5	0101	14	1110
6	0110	15	1111
7	0111		
8	1000		

The Machine START must be issued first, followed by the Robot's START.

MACHINE: ISSUE START

Press the Machine START button.

START

The MACHINE START PUSHBUTTON ACKNOWLEDGED
turns On.

MACHINE SYSTEM SIGNAL

MACHINE START PUSHBUTTON
ACKNOWLEDGED



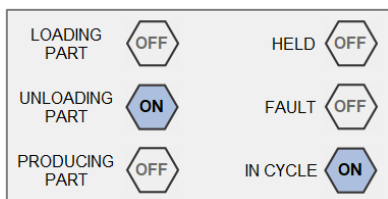
continue

ROBOT: ISSUE START

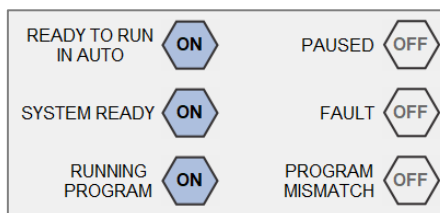
Press the Robot START button.

START

The Machine's UNLOADING PART and IN CYCLE indicators turn On.



The Robot's RUNNING PROGRAM indicators turn On.



When the Load Part cycle is finished the indicators turn Off.

Note: robot returns to a Paused status.

MACHINE FAULT

Select a Machine ISSUE LOAD or ISSUE UNLOAD.

Use the Robot Numeric Input to enter a value for the PROGRAM STYLE SELECT.

Set the Robot STYLE ACKNOWLEDGED NUMBER VALUE.

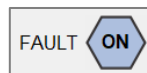
Be ready to select the FAULT switch while the program is In Cycle.

Press the Machine START and Robot START pushbuttons.

Select the Machine FAULT switch.



The Machine and Robot FAULT indicators turn On.



INFORMATION - A Fault on the Machine will cause a Fault on the Robot. For both devices the Fault aborts the programs.

Press the Machine RESET pushbutton.

RESET

The Machine FAULT indicator turns Off.

Press the Robot RESET pushbutton.

RESET

The Robot FAULT indicator turns Off.



ROBOT FAULT

Select a Machine ISSUE LOAD or ISSUE UNLOAD.

Use the Robot Numeric Input to enter a value for the PROGRAM STYLE SELECT.

Set the Robot STYLE ACKNOWLEDGED NUMBER VALUE.

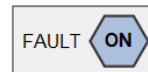
Be ready to select the FAULT switch while the program is In Cycle.

Press the Machine START and Robot START pushbuttons.

Select the Robot FAULT switch.



The Machine and Robot FAULT indicators turn On.



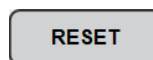
INFORMATION - A Fault on the Robot will cause a Fault on the Machine. For both devices the Robot Fault aborts the programs.

Press the Machine RESET pushbutton.



The Machine FAULT indicator turns Off.

Press the Robot RESET pushbutton.



The Robot FAULT indicator turns Off.

MACHINE HOLD

Select a Machine ISSUE LOAD or ISSUE UNLOAD.

Use the Robot Numeric Input to enter a value for the PROGRAM STYLE SELECT.

Set the Robot STYLE ACKNOWLEDGED NUMBER VALUE.

Be ready to select the HOLD pushbutton while the program is In Cycle.

Press the Machine START and Robot START pushbuttons.

Press the Machine HOLD pushbutton.



continue



The Machine HELD and Robot PAUSED indicators turn On.

INFORMATION – The Machine and Robot must coordinate their actions. A Hold on one device will stop both devices.

Press the Machine START pushbutton.

START

Press the Robot START pushbutton.

START

The Machine and Robot resume their cycles.

ROBOT HOLD

Select a Machine ISSUE LOAD or ISSUE UNLOAD.

Use the Robot Numeric Input to enter a value for the PROGRAM STYLE SELECT.

Set the Robot STYLE ACKNOWLEDGED NUMBER VALUE.

Be ready to select the HOLD pushbutton while the program is In Cycle.

Press the Machine START and Robot START pushbuttons.

Press the Robot HOLD pushbutton.

HOLD

The Machine HELD and Robot PAUSED indicators turn On.

INFORMATION – The Machine and Robot must coordinate their actions. A Hold on one device will stop both devices.

Press the Machine START pushbutton.

START

Press the Robot START pushbutton.

START

The Machine and Robot resume their cycles.

continue



INFORMATION - Design Note

The **Machine** and the **Robot** have separate **START** signals. With the HMI your pressing of the individual START buttons is similar to a “system start” signals to the two devices. An application may use a **Master Start** where the one pushbutton (or signal) is used to initiate the starting sequence to the machine and robot.

A similar design consideration could be for the individual **Machine** and **Robot RESET** signals. Using a **Master Reset** option would require a Safety Risk Assessment to determine if a Master Reset would present a hazard to the facilities personnel.

ROBOT PROGRAM MISMATCH

Select a Machine ISSUE LOAD or ISSUE UNLOAD.

Use the Robot Numeric Input to enter a value for the PROGRAM STYLE SELECT.

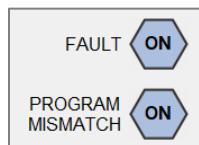
Set the Robot STYLE ACKNOWLEDGED NUMBER VALUE **that is not the same as the PROGRAM STYLE SELECT**.

SET THE BITS FOR STYLE ACKNOWLEDGED
NUMBER VALUE

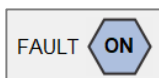
07	06	05	04	03	02	01	00
----	----	----	----	----	----	----	----

DECIMAL	BINARY	DECIMAL	BINARY
0	0000	9	1001
1	0001	10	1010
2	0010	11	1011
3	0011	12	1100
4	0100	13	1101
5	0101	14	1110
6	0110	15	1111
7	0111		
8	1000		

Press the Machine START and Robot START pushbuttons.



The Robot Program MISMATCH and FAULT indicators turn On.



The Machine FAULT indicators turn On.

INFORMATION - A Robot Program Mismatch will cause a Fault on the Robot. The robot is not running the correct program for the selected Load or Unload cycle. For both devices the Mismatch aborts the programs.

continue



Press the Robot RESET pushbutton.

RESET

The Robot FAULT and PROGRAM MISMATCH indicators turn Off.

FOR GRADE

Demonstrate the following functions:

- Part Unload Cycle
- Produce Part Cycle
- Unload Part Cycle
- Machine In Cycle Fault
- Robot Running Fault
- Machine In Cycle Hold
- Robot Running Hold
- Robot Program Mismatch



APPENDIX A: I/O ASSIGNMENTS

ROBOT INPUTS	ASSIGNMENTS	PLC OUTPUTS	TAGS
UI[1]	*IMSTP	Robot_1:O1.Output[0].0	Robot_1_UI01_IMSTP
UI[2]	*HOLD	Robot_1:O1.Output[0].1	Robot_1_UI02_HOLD
UI[3]	*SFSPD	Robot_1:O1.Output[0].2	Robot_1_UI03_SFSPD
UI[4]	*CSTOPI	Robot_1:O1.Output[0].3	Robot_1_UI04_CSTOPI
UI[5]	*FAULT RESET	Robot_1:O1.Output[0].4	Robot_1_UI05_FAULT_RESET
UI[6]	START	Robot_1:O1.Output[0].5	Robot_1_UI06_START
UI[7]	HOME	Robot_1:O1.Output[0].6	Robot_1_UI07_HOME
UI[8]	**ENBL	Robot_1:O1.Output[0].7	Robot_1_UI08_ENBL
UI[9]	STYLE1/RSR1/PNS1	Robot_1:O1.Output[1].0	Robot_1_UI09_STYLE1_RSR1_PNS1
UI[10]	STYLE2/RSR2/PNS2	Robot_1:O1.Output[1].1	Robot_1_UI10_STYLE2_RSR2_PNS2
UI[11]	STYLE3/RSR3/PNS3	Robot_1:O1.Output[1].2	Robot_1_UI11_STYLE3_RSR3_PNS3
UI[12]	STYLE4/RSR4/PNS4	Robot_1:O1.Output[1].3	Robot_1_UI12_STYLE4_RSR4_PNS4
UI[13]	STYLE5/PNS5/RSR5	Robot_1:O1.Output[1].4	Robot_1_UI13_STYLE5_PNS5_RSR5
UI[14]	STYLE6/PNS6/RSR6	Robot_1:O1.Output[1].5	Robot_1_UI14_STYLE6_PNS6_RSR6
UI[15]	STYLE7/PNS7/RSR7	Robot_1:O1.Output[1].6	Robot_1_UI15_STYLE7_PNS7_RSR7
UI[16]	STYLE8/PNS8/RSR8	Robot_1:O1.Output[1].7	Robot_1_UI16_STYLE8_PNS8_RSR8
UI[17]	PNSTROBE	Robot_1:O1.Output[2].0	Robot_1_UI17_PNSTROBE
UI[18]	PROD_START	Robot_1:O1.Output[2].1	Robot_1_UI18_PROD_START
DI[1]	DIGITAL INPUT 1	Robot_1:O1.Output[2].2	Robot_1_DI_01
DI[2]	DIGITAL INPUT 2	Robot_1:O1.Output[2].3	Robot_1_DI_02
DI[3]	DIGITAL INPUT 3	Robot_1:O1.Output[2].4	Robot_1_DI_03
DI[4]	DIGITAL INPUT 4	Robot_1:O1.Output[2].5	Robot_1_DI_04
DI[5]	DIGITAL INPUT 5	Robot_1:O1.Output[2].6	Robot_1_DI_05
DI[6]	DIGITAL INPUT 6	Robot_1:O1.Output[2].7	Robot_1_DI_06
DI[7]	DIGITAL INPUT 7	Robot_1:O1.Output[3].0	Robot_1_DI_07
DI[8]	DIGITAL INPUT 8	Robot_1:O1.Output[3].1	Robot_1_DI_08
GI[1]	GROUP INPUT 1 BIT 1	Robot_1:O1.Output[3].2	Robot_1_GI_01_Binary1 ←
GI[1]	GROUP INPUT 1 BIT 2	Robot_1:O1.Output[3].3	Robot_1_GI_01_Binary2
GI[1]	GROUP INPUT 1 BIT 3	Robot_1:O1.Output[3].4	Robot_1_GI_01_Binary4
GI[1]	GROUP INPUT 1 BIT 4	Robot_1:O1.Output[3].5	Robot_1_GI_01_Binary8
	UNUSED	Robot_1:O1.Output[3].6	UNUSED_IN1
	UNUSED	Robot_1:O1.Output[3].7	UNUSED_IN2



ROBOT OUTPUTS	ASIGNMENTS	PLC INPUTS	TAGS
UO[1]	CMDENBL	Robot_1:I1.Input[0].0	Robot_1_UO01_CMDENBL
UO[2]	SYSRDY	Robot_1:I1.Input[0].1	Robot_1_UO02_SYSRDY
UO[3]	PROGRUN	Robot_1:I1.Input[0].2	Robot_1_UO03_PROGRUN
UO[4]	PAUSED	Robot_1:I1.Input[0].3	Robot_1_UO04_PAUSED
UO[5]	HELD	Robot_1:I1.Input[0].4	Robot_1_UO05_HELD
UO[6]	FAULT	Robot_1:I1.Input[0].5	Robot_1_UO06_FAULT
UO[7]	ATPERCH	Robot_1:I1.Input[0].6	Robot_1_UO07_ATPERCH
UO[8]	TPENBL	Robot_1:I1.Input[0].7	Robot_1_UO08_TPENBL
UO[9]	BATALM	Robot_1:I1.Input[1].0	Robot_1_UO09_BATALM
UO[10]	BUSY	Robot_1:I1.Input[1].1	Robot_1_UO10_BUSY
UO[11]	STYLEACK1/ACK1/SNO1	Robot_1:I1.Input[1].2	Robot_1_UO11_STYLE1_ACK1_SNO1
UO[12]	STYLEACK2/ACK2/SNO2	Robot_1:I1.Input[1].3	Robot_1_UO12_STYLE2_ACK2_SNO2
UO[13]	STYLEACK3/ACK3/SNO3	Robot_1:I1.Input[1].4	Robot_1_UO13_STYLE3_ACK3_SNO3
UO[14]	STYLEACK4/ACK4/SNO4	Robot_1:I1.Input[1].5	Robot_1_UO14_STYLE4_ACK4_SNO4
UO[15]	STYLEACK5/SNO5/ACK5	Robot_1:I1.Input[1].6	Robot_1_UO15_STYLE5_SNO5_ACK5
UO[16]	STYLEACK6/SNO6/ACK6	Robot_1:I1.Input[1].7	Robot_1_UO16_STYLE6_SNO6_ACK6
UO[17]	STYLE7/SNO7/ACK7	Robot_1:I1.Input[2].0	Robot_1_UO17_STYLE7_SNO7_ACK7
UO[18]	STYLE8/SNO8/ACK8	Robot_1:I1.Input[2].1	Robot_1_UO18_STYLE8_SNO8_ACK8
UO[19]	SNACK	Robot_1:I1.Input[2].2	Robot_1_UO19_SNACK
UO[20]	RESERVED	Robot_1:I1.Input[2].3	Robot_1_UO20_RESERVED
DO[1]	DIGITAL OUTPUT 1	Robot_1:I1.Input[2].4	Robot_1_DO_01
DO[2]	DIGITAL OUTPUT 2	Robot_1:I1.Input[2].5	Robot_1_DO_02
DO[3]	DIGITAL OUTPUT 3	Robot_1:I1.Input[2].6	Robot_1_DO_03
DO[4]	DIGITAL OUTPUT 4	Robot_1:I1.Input[2].7	Robot_1_DO_04
DO[5]	DIGITAL OUTPUT 5	Robot_1:I1.Input[3].0	Robot_1_DO_05
DO[6]	DIGITAL OUTPUT 6	Robot_1:I1.Input[3].1	Robot_1_DO_06
DO[7]	DIGITAL OUTPUT 7	Robot_1:I1.Input[3].2	Robot_1_DO_07
DO[8]	DIGITAL OUTPUT 8	Robot_1:I1.Input[3].3	Robot_1_DO_08
GO[1]	GROUP OUTPUT 1 BIT 1	Robot_1:I1.Input[3].4	Robot_1_GO_01_Binary1
GO[1]	GROUP OUTPUT 1 BIT 2	Robot_1:I1.Input[3].5	Robot_1_GO_01_Binary2
GO[1]	GROUP OUTPUT 1 BIT 3	Robot_1:I1.Input[3].6	Robot_1_GO_01_Binary4
GO[1]	GROUP OUTPUT 1 BIT 4	Robot_1:I1.Input[3].7	Robot_1_GO_01_Binary8



APPENDIX B: FANUC USER OPERATOR PANEL ASSIGNMENTS

USER OPERATOR PANEL (UOP) INPUTS

UOP Input Signals	UOP UI
*IMSTP	UI 1
*HOLD	UI 2
*SFSPD	UI 3
*CSTOPI ⁽¹⁾	UI 4
*FAULT RESET ⁽¹⁾	UI 5
START	UI 6
HOME	UI 7
**ENBL	UI 8
STYLE1/RSR1/PNS1	UI 9
STYLE2/RSR2/PNS2	UI 10
STYLE3/RSR3/PNS3	UI 11
STYLE4/RSR4/PNS4	UI 12
STYLE5/PNS5/RSR5 ⁽²⁾	UI 13
STYLE6/PNS6/RSR6 ⁽²⁾	UI 14
STYLE7/PNS7/RSR7 ⁽²⁾	UI 15
STYLE8/PNS8/RSR8 ⁽²⁾	UI 16
PNSTROBE	UI 17
PROD_START	UI 18

* Logic 1 or ON signal required for robot operations.

⁽¹⁾ Signals require a transition to be active.

** Logic 1 or ON signal required for robot operations.

⁽²⁾ If a second Motion Group is used.

UOP Input Signal	Description
*IMSTP Held ON signal UI[1]	This input is the immediate stop software signal. *IMSTP is a signal held ON. When it is set to OFF, it • Pauses the program. • Immediately stops the robot and applies robot brakes • Shuts off power to the servos  WARNING *IMSTP is a software controlled input and cannot be used for safety purposes.
*HOLD Held ON signal UI[2]	This input is the external hold signal. *Hold is a signal held ON. When it is set to OFF, it will do the following: • Pause program execution • Slow motion to a controlled stop and hold • Does not shut off servo power.



UOP Input Signal	Description
*SFSPD Held ON signal UI[3]	This input is the safety speed input signal. This signal is usually connected to the safety fence. *SFSPD is a signal held ON. When it is set to OFF it will do the following: • Pause program execution • Reduce the speed override value to that defined in a system variable. This value cannot be increased while *SFSPD is OFF. • Display error code message SYST009. • Not allow is a REMOTE start condition for robot motion. Start inputs from UOP or SOP are disabled when SFSPD is set to OFF and only the teach pendant has motion control with the speed clamped.
*CSTOPI Held ON signal UI[4]	This input is the cycle stop input. *CSTOPI is a signal held ON. When it is set to OFF, it will do the following: The function of this signal depends on the system variable \$SHELL_CFG.\$USE_ABORT. \$SHELL_CFG.\$USE_ABORT = FALSE  WARNING \$SHELL_CFG.\$USE_ABORT = FALSE, CSTOPI does not immediately stop automatic program execution. Automatic execution will be stopped after the current program has finished executing. • Clears the queue of programs to be executed that were sent by RSR signals \$SHELL_CFG.\$USE_ABORT = TRUE • Immediately aborts the currently executing program(s) for programs that were sent to be executed by either RSR or PNS. • Clears the queue of programs to be executed that were sent by RSR signals.

continue

UOP Input Signal	Description
* FAULT_RESET Held ON signal UI[5]	This input is the external fault reset signal. When this signal transitions from ON to OFF the following will happen: • Error status is cleared • Servo power is turned on • The paused program <i>will not be resumed</i>
START (CMDENBL = ON) UI[6]	This input is the remote start input. The function of this signal depends on the system variable \$SHELL_CFG.\$CONT_ONLY. \$SHELL_CFG.\$CONT_ONLY = FALSE • Resumes a Paused program • If a program is Aborted, the currently selected program starts from the position of the cursor. \$SHELL_CFG.\$CONT_ONLY = TRUE • START resumes a Paused program only. • <i>The PROD_START input must be used to start an Aborted program from the beginning.</i>
HOME (CMDENBL = ON) UI[7]	This input is the home input. When this signal is received the robot moves to the defined home position. You configure the system to do this by setting up a macro program to run when UI[7] is received.
**ENBL Held ON signal UI[8]	This input is the enable input. This signal must be ON to have motion control ability. When this signal is OFF, robot motion cannot be done. When ENBL is ON and the REMOTE switch on the operator panel is in the REMOTE position, the robot is in a remote operating condition.

continue



UOP Input Signal	Description
STYLE 1-8 RSR 1-8 PNS 1-8 (CMDENBL = ON) UI[9] - UI[16]	Signals are Binary Numbers read as an Integer value. The inputs select the Program Number. Style or PNS selects a program for execution but <i>does not execute programs.</i> The selected Style or PNS Programs are executed using the START input or the PROD_START input. See \$SHELL_CFG.\$CONT_ONLY. RSR programs are queued and execute in order queued. Type of signal depends on Production Select Method.
PNSTROBE (CMDENBL = ON) UI[17]	PNS only, a strobe signal to read Program Number Select UI[9] – UI[16] input signals.
PROD_START Active when the robot is in a remote condition (CMDENBL = ON) UI[18]	This item is the Production Start Input when used with STYLE or PNS to initiate execution of the selected Program Number UI[9] – UI[16].

USER OPERATOR PANEL (UOP) OUTPUTS

UOP Output Signals	UOP UO
CMDENBL	UO 1
SYSRDY	UO 2
PROGRUN	UO 3
PAUSED	UO 4
HELD	UO 5
FAULT	UO 6
ATPERCH	UO 7
TPENBL	UO 8
BATALM	UO 9
BUSY	UO 10
STYLEACK1/ACK1/SNO1	UO 11
STYLEACK2/ACK2/SNO2	UO 12
STYLEACK3/ACK3/SNO3	UO 13
STYLEACK4/ACK4/SNO4	UO 14
STYLEACK5/SNO5/ACK5 ⁽¹⁾	UO 15
STYLEACK6/SNO6/ACK6 ⁽¹⁾	UO 16
STYLEACK7/SNO7/ACK7 ⁽¹⁾	UO 17
STYLEACK8/SNO8/ACK8 ⁽¹⁾	UO 18
SNACK	UO 19
RESERVED	UO 20

⁽¹⁾ If a second Motion Group is used.

UOP Output Signal	Description
CMDENBL UO[1]	This is the command enable output. This output indicates that the robot is in a remote condition and ready to run in AUTO. A Fault condition will turn OFF the output. When SYSRDY is OFF, CMDENBL is OFF. This signal goes ON when the following conditions are all satisfied: • Teach pendant disabled • Mode selection switch is set to AUTO • Robot set to Remote • *IMSTP input is ON • *HOLD input is ON • *SFSPD input is ON • *CSTOPI input is ON • **ENBL input is ON



UOP Output Signal	Description
SYSRDY UO[2]	This output is the system ready output. This output indicates that the system can provide power for the servo motors.
PROGRUN UO[3]	This output turns on when a program is running.
PAUSED UO[4]	This output is the paused program output. This output turns ON when a program is paused. If the Teach Pendant or UOP Hold is issued the robot is paused.
HELD UO[5]	This output is the hold output. This output turns on when the SOP HOLD button has been pressed, or the UOP *HOLD input is OFF.
FAULT UO[6]	This output is the error output. This output turns on when the controller/program is in an error condition.
ATPERCH UO[7]	This output is the at perch output. This output turns on when the robot reaches the predefined perch position. ATPERCH position = Reference position assigned UO[7].
TPENBL UO[8]	This output is the teach pendant enable output. This output turns on when the teach pendant is on and keyswitch is in Teach mode.
BATALM UO[9]	This output is the battery alarm output. This output turns on when the CMOS RAM battery voltage goes below 2.6 volts.
BUSY UO[10]	This output is the processor busy output. This signal turns on when the robot is executing a program or when the processor is busy.



UOP Output Signal	Description
STYLE 1-8 ACK 1-8 SNO 1-8 UO[11] - UO[18]	Signals are Binary Numbers read as an Integer value. These outputs signal the acknowledgement selected PNS/Style Program Number. Normally the acknowledgement UO[11] - UO[18] and the PNS/Style number UI[9] - UI[16] are the same number value. A mismatch means an Invalid Program Number Read.
SNACK UO[19]	This output is turned (ON) as a signal to read the PNS/Style program number acknowledge outputs
RESERVED UO[20]	Unassigned

APPENDIX C: BINARY NUMBERING

To determine the equivalent DECIMAL number or BINARY number, use the following procedure.

Use the following chart of the “weighed value” for each binary digit position.

$$\begin{array}{cccc} 2^3 = 8 & 2^2 = 4 & 2^1 = 2 & 2^0 = 1 \\ \hline 8 & 4 & 2 & 1 \end{array}$$

Assign the binary number **0011** to each of the corresponding “weighed value” position.

$$\begin{array}{cccc} \hline 8 & 4 & 2 & 1 \\ \hline 0 & 0 & 1 & 1 \end{array}$$

Then, multiply the binary digit (0 or 1) by the “weighed value” position.

$$\begin{array}{cccc} \hline 8 & 4 & 2 & 1 \\ \hline 0 & 0 & 1 & 1 \\ 0 & 0 & 2 & 1 \end{array}$$

Finally, add the resulting values for the equivalent decimal number, **3**.

$$\begin{array}{cccc} \hline 8 & 4 & 2 & 1 \\ \hline 0 & 0 & 1 & 1 \\ 0 + & 0 + & 2 + & 1 = 3 \end{array}$$

Example 1:

$$\begin{array}{cccc} \hline 8 & 4 & 2 & 1 \\ \hline 0 & 1 & 0 & 1 \\ 0 + & 4 + & 0 + & 1 = 5 \end{array}$$

Example 2:

$$\begin{array}{cccc} \hline 8 & 4 & 2 & 1 \\ \hline 1 & 0 & 1 & 0 \\ 8 + & 0 + & 2 + & 0 = 10 \end{array}$$

